



Transitioning to Sustainable Urban Mobility: Behavioral and Economic Drivers of Solar-Assisted Micro-Mobility Adoption in Emerging Urban Indonesia

Edi Purwanto^{1*}, Hari Nugraha^{2*}, Ismail Alif Siregar²

¹ Department of Management, Universitas Pembangunan Jaya, 15413 South Tangerang, Indonesia

² Department of Product Design, Universitas Pembangunan Jaya, 15413 South Tangerang, Indonesia

* Correspondence: Edi Purwanto (edi.purwanto@upj.ac.id); Hari Nugraha (hari.nugraha@upj.ac.id)

Received: 05-02-2026

Revised: 07-13-2026

Accepted: 07-15-2026

Citation: Purwanto, E., Nugraha, H., & Siregar, I. A. (2026). Transitioning to sustainable urban mobility: Behavioral and economic drivers of solar-assisted micro-mobility adoption in emerging urban Indonesia. *Chall. Sustain.*, 14(4), 714–741. <https://doi.org/10.56578/cis140406>.



© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: The transition toward sustainable urban mobility in emerging economies requires not only technological innovation but also a deeper understanding of market adoption dynamics. This study examines the behavioral determinants of the adoption of solar-assisted micro-mobility among micro, small, and medium enterprises (MSMEs) in urban Indonesia. Positioned at the intersection of renewable energy integration and sustainable transportation systems, the research investigates how environmental concern, performance expectancy, operational cost, and price shape users' attitudes and subsequently influence adoption intention. A quantitative approach was employed using survey data collected from 300 MSME operators, and the relationships among constructs were analyzed using Partial Least Squares Structural Equation Modeling (PLS-SEM). The results indicate that attitude serves as a central mediating mechanism through which environmental, technological, and economic factors are significantly associated with adoption intention. Among the predictors, performance expectancy emerges as the most influential driver, followed by environmental concern, while operational cost and price play supporting but significant roles within a value-based evaluation framework. The findings highlight that adoption of solar-assisted mobility is primarily driven by perceived functional benefits and sustainability value rather than cost considerations alone. This study contributes to the sustainability literature by advancing a holistic behavioral adoption model for renewable energy-based mobility in emerging urban contexts. The results provide practical implications for policymakers and industry stakeholders in designing strategies to accelerate the diffusion of low-carbon mobility solutions, particularly among resource-constrained MSMEs.

Keywords: Sustainable urban mobility; Solar-assisted micro-mobility; Renewable energy integration; Micro, small, and medium enterprises; Technology adoption; Behavioral intention; Partial Least Squares Structural Equation Modeling; Emerging economies

1. Introduction

The rapid growth of urban economies in developing countries has significantly increased the demand for efficient and sustainable mobility solutions (Kumar & Alok, 2020). In Indonesia, particularly in metropolitan regions such as Jakarta, Bogor, Depok, Great Tangerang, and Bekasi, urbanization has intensified transportation challenges, including traffic congestion, rising greenhouse gas emissions, and heavy reliance on fossil fuels (Yang et al., 2026). These issues not only threaten environmental sustainability but also directly affect the operational efficiency of micro, small, and medium enterprises (MSMEs), which depend on flexible, low-cost transportation systems for daily business activities (Pereira et al., 2025). Empirical evidence indicates that urban transportation is a major contributor to environmental degradation while simultaneously shaping economic productivity at the micro-enterprise level (Hardman et al., 2018).

Electric vehicles (EVs) have emerged as a promising solution to mitigate these challenges by reducing emissions and improving energy efficiency (Kumar & Alok, 2020; Rezvani et al., 2015). However, despite global advancements, EV adoption among MSMEs in developing economies remains relatively low (Luthra et al., 2015).

High initial investment costs, limited charging infrastructure, and uncertainty regarding operational performance continue to hinder widespread adoption. For MSMEs, which typically operate under tight financial constraints, the decision to adopt new mobility technologies is strongly influenced by economic feasibility, reliability, and immediate business benefits rather than environmental considerations alone (Hackbarth & Madlener, 2013).

In response to these limitations, hybrid solar-electric mobility solutions have been introduced as an alternative innovation pathway. The Hybrid Solar-Electric Cart (HSEC), as a prototype technology, integrates photovoltaic energy systems with electric propulsion, offering a cost-efficient, low-emission, and infrastructure-independent transport solution. This hybrid configuration allows partial energy self-sufficiency, reducing reliance on external charging systems while maintaining operational flexibility. Such characteristics make the technology particularly relevant for MSMEs operating in dense urban environments with limited access to charging infrastructure.

While hybrid solar-electric mobility shares certain characteristics with conventional EVs, its adoption context differs fundamentally. Unlike passenger EVs, which are primarily acquired for personal transportation, the HSEC is intended as a productive business asset supporting income-generating activities among MSMEs. Consequently, adoption decisions are not merely consumer choices but operational investment decisions in which business productivity, operational continuity, environmental sustainability, and financial feasibility are evaluated simultaneously. This distinction suggests that the behavioral mechanisms underlying HSEC adoption may differ from those commonly reported in the broader EV adoption literature, where private consumption motives dominate. By positioning renewable-energy-assisted micro-mobility within an MSME operational context, this study extends the discussion of sustainable transportation beyond household mobility toward business-oriented urban mobility systems.

Despite the growing body of research on electric mobility adoption, several important theoretical and empirical gaps remain. First, previous studies have predominantly examined passenger EVs, private consumers, or commercial fleet operators, whereas renewable-energy-assisted micro-mobility designed for MSMEs has received very limited scholarly attention. As a result, existing behavioral adoption models have been developed largely within consumer-oriented contexts and may not adequately explain technology adoption decisions made for productive business purposes.

Second, MSMEs evaluate transportation technologies differently from household consumers. For mobile and semi-mobile business operators, mobility functions simultaneously as a production resource, a logistics instrument, and a source of competitive advantage. Therefore, adoption decisions are expected to integrate environmental considerations with operational performance, business continuity, and economic efficiency. This operational decision-making perspective remains largely absent from the current sustainable mobility literature.

Third, previous studies generally investigate renewable mobility technologies after they have become commercially available. Comparatively little attention has been devoted to understanding behavioral intention toward prototype-based renewable-energy innovations before market commercialization. Examining purchase intention at this pre-commercialization stage is particularly important because technological feasibility alone does not guarantee market acceptance.

Finally, although attitude has frequently been incorporated into technology adoption models, relatively limited research has investigated its role as a psychological mechanism through which environmental, technological, and economic evaluations are transformed into purchase intention within business-oriented renewable mobility systems operating in emerging economies.

From a theoretical perspective, this study argues that renewable-energy-assisted micro-mobility adoption among MSMEs represents a distinct behavioral setting compared with conventional consumer technology adoption. Because the technology functions as an operational business resource rather than solely as a transportation product, users are expected to evaluate sustainability attributes alongside productivity enhancement, operational efficiency, and long-term economic value. Accordingly, the present study positions HSEC adoption within a business-oriented technology adoption perspective, thereby extending the applicability of established behavioral adoption theories to an emerging renewable mobility context characterized by resource-constrained entrepreneurial decision-making.

Based on the identified gaps, the central problem addressed in this study is the limited understanding of how MSME actors in urban Indonesia form purchase intentions toward hybrid solar-electric mobility innovations. Specifically, it remains unclear how environmental, technological, and economic factors interact to shape attitudes and ultimately influence behavioral intention in a context characterized by resource constraints and infrastructural limitations.

Although hybrid solar-electric technologies offer clear advantages in terms of sustainability and operational efficiency, their adoption cannot be assumed without a comprehensive understanding of market perception. The absence of empirical evidence on the behavioral drivers of adoption creates uncertainty for policymakers, manufacturers, and innovators seeking to promote such technologies. Therefore, a systematic investigation is required to identify the key determinants influencing purchase intention and to understand the underlying psychological mechanisms driving adoption decisions.

This study aims to examine the behavioral determinants associated with MSMEs' intention to adopt a HSEC in urban Indonesia by integrating environmental, technological, and economic perspectives within a business-

oriented technology adoption framework. Using Partial Least Squares Structural Equation Modeling (PLS-SEM), the study evaluates how environmental concern, performance expectancy, operational cost, and price are associated with attitude and subsequently with purchase intention toward a renewable-energy-assisted mobility innovation designed for productive business use.

This study makes three principal contributions to the sustainable mobility literature. First, it introduces renewable-energy-assisted micro-mobility for MSMEs as a distinct empirical context that differs conceptually from conventional consumer EV adoption. By emphasizing productive business mobility rather than private transportation, the study expands the scope of sustainable mobility research toward enterprise-oriented applications.

Second, the study extends behavioral technology adoption literature by demonstrating that established constructs, including environmental concern, performance expectancy, operational cost, price, and attitude, operate within a different decision-making environment where mobility technologies function simultaneously as operational assets and sustainability innovations. This perspective provides a richer understanding of technology adoption among resource-constrained MSMEs in emerging economies.

Third, the study contributes to the commercialization literature on renewable-energy innovation by providing empirical evidence regarding market acceptance at the prototype stage. Rather than examining an already commercialized technology, this research investigates behavioral intention before market diffusion, thereby offering practical insights for innovation commercialization, product development, and sustainable urban mobility policy.

By integrating sustainable transportation, renewable-energy innovation, and business-oriented technology adoption, this study seeks to bridge the gap between technological feasibility and market acceptance, thereby contributing to a more comprehensive understanding of sustainable urban mobility transitions in emerging economies.

2. Literature Review

2.1 Solar-Assisted Micro-Mobility and Sustainable Urban Transportation

The rapid urbanization of developing economies has intensified the need for sustainable and efficient transportation systems, particularly in densely populated metropolitan regions. Urban transportation is widely recognized as a major contributor to greenhouse gas emissions and environmental degradation, making sustainable mobility a central component of low-carbon urban development. Electric mobility has emerged as a key solution, offering significant potential to reduce emissions and improve energy efficiency in urban transport systems (Figueiredo & Baptista, 2025).

However, the adoption of conventional EVs faces several structural barriers, particularly in emerging economies. These include high initial investment costs, limited charging infrastructure, and uncertainty regarding technological reliability. These constraints are especially pronounced among MSMEs, which typically operate under financial and operational limitations. As a result, alternative mobility solutions that reduce dependence on centralized infrastructure have become increasingly relevant.

In this context, solar-assisted micro-mobility represents an emerging innovation pathway that integrates renewable energy with localized transport systems. Hybrid solar-electric mobility solutions enable partial energy self-sufficiency by utilizing photovoltaic systems to supplement electric propulsion. This decentralized energy approach aligns with broader trends in distributed renewable energy systems, which have been shown to enhance accessibility, reduce operational dependency on centralized grids, and support inclusive economic development (Scheja & Kim, 2024).

From a sustainability perspective, solar-assisted micro-mobility offers a dual advantage. First, it contributes to environmental sustainability by reducing reliance on fossil fuels and lowering emissions. Second, it enhances economic sustainability by providing cost-efficient mobility solutions tailored to small-scale business operations. These characteristics make hybrid solar-electric mobility particularly relevant for MSMEs operating in urban environments with infrastructure constraints.

Despite its potential, empirical research on solar-assisted micro-mobility remains limited. Most existing studies focus on passenger EVs or large-scale transport systems, leaving a significant gap in understanding the adoption dynamics of small-scale, renewable-energy-based mobility innovations. Therefore, examining solar-assisted micro-mobility within a behavioral and market readiness framework is essential to bridge the gap between technological innovation and real-world adoption.

Unlike conventional passenger EVs, solar-assisted micro-mobility for MSMEs is embedded within productive economic activities rather than personal transportation. Consequently, mobility decisions are evaluated not only in terms of environmental sustainability but also in terms of business continuity, revenue generation, operational flexibility, and logistics efficiency. This distinction suggests that the behavioral logic underlying technology adoption among MSMEs differs fundamentally from household vehicle adoption. Sustainable transportation in

this context therefore represents an intersection between environmental transition and small-business competitiveness, providing a theoretically distinct setting for examining technology adoption.

2.2 Hybrid Solar-Electric Cart as a Market-Oriented Renewable Energy Innovation

The HSEC represents a novel form of renewable-energy-based micro-mobility that integrates photovoltaic energy systems with electric propulsion. Unlike conventional EVs, which rely heavily on external charging infrastructure, the HSEC offers a hybrid energy configuration that allows partial independence from grid-based energy systems. This feature is particularly advantageous in urban environments where charging infrastructure is limited or unevenly distributed.

From an innovation perspective, the HSEC can be classified as a market-oriented technological innovation, where technical feasibility must be complemented by user acceptance and economic viability. The diffusion of such innovations depends not only on engineering performance but also on how potential users perceive the technology in terms of usefulness, affordability, and relevance to their daily activities. This aligns with broader research on renewable energy adoption, which emphasizes that technological advancement alone does not guarantee market acceptance.

From an innovation diffusion perspective, the HSEC differs from conventional EVs because its value proposition extends beyond transportation (Uddin et al., 2024). For MSMEs, the HSEC functions simultaneously as a mobility platform, an income-generating asset, and an operational production resource. Consequently, adoption is expected to depend not only on perceived technological superiority but also on whether the innovation supports business productivity while reducing operational uncertainty (Purwanto et al., 2024).

This business-oriented positioning also distinguishes HSEC commercialization from consumer-oriented green technologies. Whereas private consumers often evaluate EVs through lifestyle preferences, environmental values, or personal mobility convenience, MSME operators are more likely to evaluate technology through return-on-investment considerations, operational reliability, and long-term business sustainability. Therefore, market acceptance of HSEC should be understood as organizationally embedded decision-making occurring at the micro-enterprise level rather than purely individual consumer behavior.

Studies on green technology adoption consistently highlight the importance of perceived value in shaping adoption decisions. For instance, perceived usefulness, environmental benefits, and economic advantages have been shown to significantly influence consumer attitudes and behavioral intentions toward sustainable technologies (Zhao et al., 2024). In the context of EV adoption, performance-related attributes and perceived benefits are critical in determining whether users are willing to transition from conventional technologies.

Furthermore, renewable energy technologies often face a “commercialization gap,” where promising prototypes fail to achieve widespread adoption due to insufficient understanding of market dynamics. This gap is particularly relevant for MSMEs, where adoption decisions are highly pragmatic and influenced by immediate operational benefits rather than long-term environmental considerations.

Therefore, positioning the HSEC as a market-oriented innovation requires an integrated understanding of technological, economic, and behavioral factors. It is not sufficient to demonstrate that the technology works; it must also be perceived as useful, cost-effective, and aligned with the operational needs of potential users. This perspective underscores the importance of incorporating behavioral models into the study of renewable energy-based mobility innovations.

2.3 Behavioral Intention and Green Technology Adoption

Understanding the adoption of renewable-energy-based mobility innovations requires a behavioral perspective that explains how individuals form intentions to adopt new technologies. Behavioral intention is widely recognized as the most immediate predictor of actual behavior, making it a central construct in the study of technology adoption.

Several theoretical frameworks have been developed to explain technology adoption behavior. The Theory of Planned Behavior (TPB), the Technology Acceptance Model (TAM), and integrated models combining behavioral and technological factors have been extensively used to analyze adoption decisions in the context of EVs and sustainable technologies (Figueiredo & Baptista, 2025). These frameworks emphasize the role of cognitive evaluations, perceived usefulness, social influences, and control factors in shaping behavioral intention.

A key construct within these models is attitude, which represents an individual’s overall evaluation of a technology. Empirical studies consistently show that attitude plays a central role in determining adoption intention. For example, research on EV adoption indicates that positive attitudes toward the technology significantly increase the likelihood of purchase intention and actual adoption behavior (Stockkamp et al., 2021). Similarly, consumer awareness and perceived benefits have been found to positively influence attitude, which in turn affects intention to adopt sustainable technologies (Lashari et al., 2021).

Moreover, recent studies highlight that behavioral intention toward green technologies is influenced by a

combination of environmental, technological, and economic factors. Environmental concern reflects individuals' awareness of ecological issues and their willingness to support sustainable solutions. Performance expectancy captures the perceived usefulness and functional benefits of the technology. Economic factors such as operational cost and price represent the financial considerations that influence adoption decisions. These multidimensional drivers interact to shape attitude, which then acts as a mediator between perception and intention.

Although behavioral adoption theories such as the TPB, TAM, and Unified Theory of Acceptance and Use of Technology (UTAUT) have been extensively validated across diverse technological contexts, most empirical applications focus on household consumers or organizational employees. Comparatively little attention has been devoted to micro-enterprises that simultaneously act as individual decision-makers and as business organizations. This dual role introduces a more complex decision-making process in which personal attitudes are closely intertwined with operational and economic considerations. Consequently, established behavioral adoption theories require contextual extension when applied to renewable-energy-assisted mobility designed for MSMEs.

Importantly, the role of attitude as a mediating variable has gained increasing attention in recent research. Rather than acting solely as a direct predictor, attitude serves as a psychological mechanism that translates perceptions into behavioral intention. This mediating role is particularly relevant in the context of emerging technologies, where users must first develop a favorable evaluation before forming a concrete intention to adopt.

2.4 Environmental Concern and Attitude Toward Hybrid Solar-Electric Cart

Environmental concern reflects the degree to which individuals are aware of environmental issues and are willing to support environmentally sustainable solutions. In the context of green technology adoption, environmental concern has been widely recognized as a significant psychological driver influencing individuals' evaluation of eco-friendly innovations.

Prior studies demonstrate that individuals with higher environmental awareness tend to develop more favorable attitudes toward low-emission technologies, including EVs and renewable energy systems. For instance, research in sustainable mobility shows that environmental concern significantly enhances positive evaluations of EVs as a means to mitigate climate change and urban pollution (Buhmann et al., 2024; Rezvani et al., 2015). Similarly, findings from the studies confirm that environmental consciousness positively shapes attitudes toward green technologies by aligning product attributes with pro-environmental values (Stockkamp et al., 2021).

However, in emerging economies, environmental concern often interacts with economic rationality. MSMEs may acknowledge environmental benefits, but their attitudes are strengthened only when such benefits are accompanied by practical utility and cost efficiency. In the case of the HSEC, environmental concern is expected to positively influence users' attitudes by reinforcing the perceived ecological value of solar-assisted mobility.

Within household EV adoption, environmental concern often reflects personal ecological values and environmental responsibility. However, among MSMEs, environmental concern may operate differently because environmental commitment is evaluated alongside business viability. Entrepreneurs are unlikely to adopt environmentally friendly technologies solely because they reduce emissions; instead, environmental value must be compatible with operational performance and economic sustainability. Accordingly, environmental concern is expected to strengthen favorable evaluations of HSEC only when the technology is simultaneously perceived as supporting business operations.

H1: Environmental concern positively influences attitude toward the HSEC.

2.5 Performance Expectancy and Attitude Toward Hybrid Solar-Electric Cart

Performance expectancy refers to the extent to which individuals believe that a technology will enhance their performance or productivity. Within the UTAUT, performance expectancy is consistently identified as a primary determinant of technology adoption. Empirical evidence from the EV literature shows that perceived usefulness, reliability, and functional performance significantly influence users' attitudes and adoption decisions (Jansson et al., 2017; Li et al., 2016; Purwanto & Irawan, 2024). Users are more likely to develop favorable attitudes when they perceive that the technology can improve efficiency, reduce operational constraints, and deliver tangible benefits.

For MSMEs, performance expectancy is particularly critical because mobility solutions directly affect business operations. The HSEC must be perceived as reliable, efficient, and capable of supporting daily business activities. Therefore, performance-related perceptions are expected to be the strongest predictor of attitude formation.

Performance expectancy is expected to play a more dominant role among MSMEs than among conventional household consumers because mobility directly affects productive business activities. Delays, reduced carrying capacity, limited operational range, or unreliable performance may directly influence daily income generation. Therefore, expected performance is not merely a convenience factor but an operational necessity. This theoretical distinction provides a strong rationale for expecting performance expectancy to become the strongest antecedent of attitude toward HSEC adoption.

H2: Performance expectancy positively influences attitude toward the HSEC.

2.6 Operational Cost and Attitude Toward Hybrid Solar-Electric Cart

Operational cost represents the perceived expenses associated with using a technology over time, including energy consumption, maintenance, and repair costs. In the context of MSMEs, operational cost is a crucial consideration because business sustainability is closely linked to cost efficiency. Research on electric mobility adoption indicates that lower operating costs significantly enhance users' attitudes and adoption intentions (Hackbarth & Madlener, 2013; Mesquita et al., 2025). Electric and hybrid vehicles are often perceived as economically advantageous due to reduced fuel consumption and lower maintenance requirements compared to conventional vehicles.

The HSEC offers potential cost advantages through reduced reliance on fossil fuels and partial energy self-sufficiency. These characteristics are expected to positively influence users' attitudes by enhancing the perceived economic value of the technology.

For resource-constrained MSMEs, operational expenditure constitutes a recurring component of business profitability. Unlike household transportation decisions, reductions in fuel consumption and maintenance expenses directly affect operating margins. Consequently, operational cost should be interpreted not merely as a financial attribute of the vehicle but as a determinant of enterprise sustainability. This perspective differentiates MSME-oriented mobility adoption from conventional consumer EV studies.

H3: Operational cost positively influences attitude toward the HSEC.

2.7 Price and Attitude Toward Hybrid Solar-Electric Cart

Price reflects the perceived initial investment required to acquire a technology. Unlike operational cost, which focuses on long-term usage, price represents the upfront financial commitment and is often a critical barrier in emerging markets. Studies in technology adoption consistently show that perceived price fairness and affordability significantly influence attitudes and purchase decisions (Egbue & Long, 2012; Li et al., 2016). High upfront costs can discourage adoption, even when long-term benefits are substantial. Conversely, when users perceive the price as reasonable relative to the expected benefits, they are more likely to form a positive attitude. In the case of the HSEC, price is expected to influence attitude by shaping users' perception of affordability and investment feasibility. This is particularly relevant for MSMEs, which often operate under financial constraints.

Although purchase price is widely recognized as an important determinant of technology adoption, MSME investment decisions frequently involve broader value assessments rather than simple cost minimization. Entrepreneurs may accept relatively higher acquisition costs when the technology is expected to improve operational efficiency, reduce long-term expenditures, and increase business productivity. Therefore, price is conceptualized within this study as one component of overall value evaluation rather than an isolated financial barrier.

H4: Price positively influences attitude toward the HSEC.

2.8 Attitude and Intention to Purchase Hybrid Solar-Electric Cart

Attitude is defined as an individual's overall evaluation of a technology, encompassing both cognitive beliefs and affective responses. In behavioral intention models such as TPB and TAM, attitude is a key predictor of intention and subsequent behavior. Empirical studies consistently confirm that a positive attitude toward green technologies significantly increases the likelihood of adoption (Megha, 2024; Paul et al., 2016). In the context of EVs, users with favorable attitudes are more likely to express an intention to purchase and adopt the technology. In this study, attitude toward the HSEC is expected to be the primary determinant of purchase intention. As users evaluate the technology based on environmental, technological, and economic considerations, their overall attitude becomes the decisive factor influencing their adoption decision.

In the present study, attitude is conceptualized as a business-oriented evaluative judgment rather than merely an affective preference toward green technology. Because the HSEC functions as an operational business asset, attitude reflects respondents' integrated assessment of environmental benefits, expected operational performance, economic efficiency, and investment feasibility. This broader conceptualization positions attitude as a cognitive integration mechanism linking multiple dimensions of evaluation to subsequent purchase intention.

H5: Attitude positively influences intention to purchase the HSEC.

2.9 Mediating Role of Attitude in Purchase Intention

Recent advancements in technology adoption research emphasize the importance of mediating mechanisms that explain how perceptions are translated into behavioral intention. Attitude is increasingly recognized as a central

mediator linking cognitive evaluations to behavioral outcomes. Studies have shown that factors such as perceived usefulness, environmental concern, and economic benefits often influence purchase intention indirectly through attitude (Wang et al., 2016). This suggests that individuals do not directly translate perceptions into intention; instead, they first form an overall evaluation of the technology, which then drives their behavioral decision.

Previous studies provide empirical support for the mediating role of attitude in green technology and EV adoption. In the context of EV adoption, Jaiswal et al. (2021) found that attitude partially mediates the effects of perceived usefulness and perceived ease of use on adoption intention, confirming that cognitive evaluations of technology are translated into intention through users' favorable evaluation of the vehicle. Similarly, a study showed that attitude mediates the relationship between perceived usefulness and adoption intention, highlighting the importance of attitude as a psychological mechanism in EV decision-making (Chanda et al., 2024).

Prior studies indicate that environmental awareness and ecological consciousness influence EV adoption through evaluative mechanisms such as attitude, perceived value, or resistance attitude. Wu et al. (2021) found that ecological consciousness promotes EV purchase intention, while value perceptions and resistant attitude mediate this relationship. Other study further demonstrated that environmental awareness significantly predicts EV adoption and contributes to perceived value formation, which subsequently leads to adoption intention (Wu et al., 2021). These findings support the argument that environmental concern does not automatically become purchase intention; rather, it must first be internalized into a favorable evaluation of the technology.

Studies grounded in technology acceptance and UTAUT perspectives show that performance-related beliefs are among the most important antecedents of EV adoption. Higuera-Castillo et al. (2023) examined EV adoption intention across India and Spain and included performance expectancy as a key determinant of adoption intention. Other study also extended the UTAUT framework in the EV adoption context and incorporated performance expectancy as a critical determinant of adoption (Ajao et al., 2025). More directly, Jaiswal et al. (2021) and Chanda et al. (2024) found that perceived usefulness influences adoption intention through attitude, providing strong support for the mediating logic proposed in H6b.

Previous studies emphasize that operational cost and perceived economic value are important in shaping attitudes and adoption intentions toward EVs. Research on EV adoption consistently identifies operating cost, perceived savings, and economic value as central considerations for potential adopters. A study found that perceived sacrifices, including cost-related concerns, influence perceived value, which then affects EV adoption intention (Mustafa et al., 2026). Other study also highlight operating costs as an important economic factor in EV ownership dynamics (Gutjar et al., 2025). These studies support the assumption that perceived operational cost efficiency strengthens purchase intention when it contributes to a more favorable evaluation of the technology.

Prior research shows that purchase price and affordability remain major barriers or determinants in EV adoption. The study found that EV purchase cost is a barrier to adoption, while positive emotions and environmental concern support adoption behavior (Salari, 2022). Another study reported that stakeholders identify purchase price as a determinant in EV purchase decisions (D'Adamo et al., 2023). Similarly, a study found that although attitudes toward EVs may be positive, relatively high purchase price remains a major issue in sustainable mobility adoption (Balassa & Koteczki, 2026). These findings support H6d by indicating that price affects purchase intention not only as a financial constraint, but also through its influence on users' evaluative attitude toward the technology.

In the context of the HSEC, environmental, technological, and economic factors are expected to influence purchase intention through attitude. This mediating role is particularly important for emerging technologies, where users must develop confidence and positive evaluation before forming adoption intentions.

The mediating role of attitude is expected to become particularly important in prototype-based renewable-energy innovations because prospective adopters have little or no direct experience with the technology. Under such conditions, objective technical information alone is insufficient to generate purchase intention. Instead, individuals first synthesize environmental, technological, and economic evaluations into an overall attitudinal judgment before considering adoption. This cognitive integration process may be even more pronounced among MSMEs because purchasing decisions involve business investment rather than routine consumer consumption. Accordingly, attitude is theorized to function as the principal psychological mechanism through which diverse perceptions are translated into purchase intention.

H6a: Attitude mediates the relationship between environmental concern and intention to purchase.

H6b: Attitude mediates the relationship between performance expectancy and intention to purchase.

H6c: Attitude mediates the relationship between operational cost and intention to purchase.

H6d: Attitude mediates the relationship between price and intention to purchase.

2.10 Research Framework and Theoretical Positioning

Collectively, the preceding literature suggests that renewable-energy-assisted micro-mobility adoption among MSMEs constitutes a theoretically distinct behavioral context. While the study adopts constructs that have been extensively examined within technology adoption literature, their interaction is expected to differ because the technology functions simultaneously as a sustainability innovation and an operational business resource.

Consequently, the proposed framework positions environmental concern, performance expectancy, operational cost, and price as complementary dimensions influencing business-oriented evaluative judgment (attitude), which subsequently predicts purchase intention (see Figure 1). This framework extends previous consumer-oriented EV adoption studies by emphasizing enterprise-level decision-making within resource-constrained urban environments.

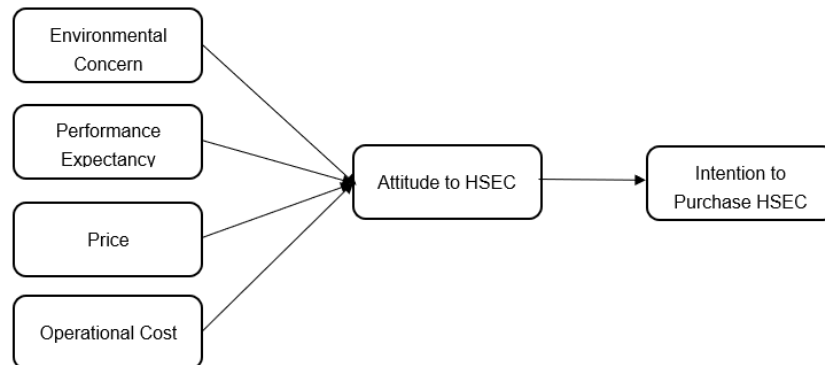


Figure 1. Conceptual framework
Note: HSEC = Hybrid Solar-Electric Cart.

3. Methodology

3.1 Prototype Description of the Hybrid Solar-Electric Cart

The object evaluated in this study is a prototype HSEC specifically developed to support sustainable urban mobility among MSMEs. The prototype integrates photovoltaic energy generation with battery-powered electric propulsion to reduce dependence on fossil fuels while maintaining sufficient operational performance for daily commercial activities. As illustrated in Figure 2, the HSEC integrates photovoltaic panels with battery-powered electric propulsion to support sustainable urban mobility.



Figure 2. Prototype of the Hybrid Solar-Electric Cart (HSEC)

Unlike conventional electric motorcycles that depend entirely on external charging infrastructure, the HSEC incorporates roof-mounted photovoltaic panels that continuously recharge the battery during daylight operation. This hybrid energy configuration extends operational range, reduces electricity consumption from the grid, and improves energy self-sufficiency during routine business activities.

The prototype was designed primarily for mobile and semi-mobile MSME operators such as street food vendors, beverage sellers, traditional market traders, and other small businesses requiring daily transportation of goods within urban environments. The vehicle emphasizes low operating cost, minimal maintenance requirements, and environmental sustainability while maintaining adequate payload capacity for commercial use.

Table 1 summarizes the principal technical specifications of the HSEC prototype presented to respondents during the survey. The prototype was designed to provide an environmentally sustainable and economically viable mobility solution for mobile and semi-mobile MSMEs operating in urban environments. Its compact dimensions, pedal-assisted electric propulsion, adequate payload capacity, and relatively long driving range make the HSEC

suitable for daily commercial activities while reducing dependence on fossil fuels. During the survey, respondents were provided with standardized photographs, video, and technical specifications of the prototype to ensure a consistent understanding of the technology before completing the questionnaire.

Table 1. Technical specifications of the Hybrid Solar-Electric Cart (HSEC)

Specification	Description
Vehicle Type	HSEC
Overall Dimensions	Length: 200 cm × Width: 70 cm × Height: 150 cm
Propulsion System	Electric motor with pedal-assisted drive
Battery Type	Lithium-ion battery
Battery Capacity	64 V, 22.5 Ah
Maximum Payload	160 kg
Capacity	
Maximum Speed	35 km/h
Driving Range	Up to 60 km per full battery charge
Battery Charging Time	Approximately 5 hours (full charge)
Primary Energy Source	Battery-powered electric propulsion supported by roof-mounted photovoltaic (solar) panels
Intended Users	Mobile and semi-mobile Micro, Small, and Medium Enterprises (MSMEs)
Intended Applications	Street food vendors, beverage sellers, traditional market traders, and other mobile urban businesses

Table 2 shows that the HSEC offers several practical advantages compared with conventional motorcycle commonly used by MSMEs. Although both vehicles require an initial investment, the HSEC has a lower estimated purchase price of IDR 20,000,000 compared with IDR 25,000,000 for conventional motorcycle. In addition, the HSEC uses electricity and solar energy, resulting in lower energy costs, zero tailpipe emissions, lower noise levels, and reduced environmental impact. Its 160 kg payload capacity and 60 km driving range indicate that the prototype is practically suitable for daily mobile and semi-mobile MSME activities. The comparison suggests that the HSEC may provide a more sustainable and cost-efficient mobility alternative for urban MSMEs, particularly for businesses seeking to reduce fuel dependence and operating expenses.

Table 2. Comparison between the Hybrid Solar-Electric Cart (HSEC) and conventional mobility used by micro, small, and medium enterprises (MSMEs)

Feature	HSEC	Conventional Motorcycle
Purchase Price	IDR 20,000,000	IDR 25,000,000
Energy Source	Electricity + Solar Energy	Gasoline
Fuel/Energy Cost	Low	Moderate to High
Tailpipe Emissions	None	High
Maximum Payload	160 kg	Lower for cargo transport
Driving Range	60 km	Depends on fuel tank
Maintenance Cost	Low	Moderate
Noise Level	Low	High
Environmental Impact	Low-carbon mobility	Fossil-fuel dependent

3.2 Research Design

This study employed a quantitative, cross-sectional research design to investigate the behavioral factors associated with MSMEs' intention to adopt the HSEC in urban Indonesia. Because the HSEC remains at the prototype stage of technological development, the research focused on measuring prospective purchase intention rather than actual adoption behavior.

The conceptual model integrates environmental, technological, and economic factors influencing attitude and subsequent purchase intention. PLS-SEM was selected because the study emphasizes prediction-oriented analysis, simultaneously estimates multiple latent constructs, and is well suited for evaluating behavioral models involving emerging technologies.

3.3 Population and Sampling Technique

The target population of this study consists of MSME actors engaged in mobile or semi-mobile business activities in urban areas, particularly in the Jakarta, Bogor, Depok, Tangerang, and Bekasi region. These include street vendors and informal business operators who rely heavily on flexible transportation systems for daily operations, such as food vendors (e.g., satay sellers, meatball vendors, fried rice vendors), beverage sellers, and other itinerant traders.

Given the absence of a comprehensive sampling frame for MSMEs operating in informal and mobile sectors, this study employed a non-probability sampling technique, specifically convenience sampling. Respondents were selected based on their accessibility, availability, and willingness to participate in the survey. This approach is widely used in exploratory and behavioral research, particularly when the population is difficult to enumerate or lacks formal registration. A total of 300 valid responses were collected, which satisfies the recommended sample size requirements for PLS-SEM analysis. The sample size is considered adequate for ensuring statistical power and model stability, especially given the number of constructs and indicators included in the study.

3.4 Data Collection Procedure

Data were collected through a structured questionnaire distributed directly to respondents in selected urban areas. The survey was conducted through face-to-face interactions to ensure that respondents clearly understood the concept of the HSEC, given that it is a prototype technology not yet widely available in the market. Before completing the questionnaire, respondents were provided with a brief explanation and visual illustration of the HSEC concept, including its key features such as solar-assisted energy systems, electric propulsion, and intended use for MSME mobility. This step was necessary to ensure that responses were based on an informed understanding of the innovation.

Before questionnaire administration, respondents received a standardized five-minute explanation regarding the HSEC. The explanation included: (1) standardized photographs; (2) a short demonstration video illustrating how the HSEC operates in real-world business activities; (3) technical specifications; (4) operational characteristics; (5) expected business applications; (6) advantages; and (7) current prototype limitations. The demonstration video was presented uniformly to all respondents to ensure a consistent understanding of the vehicle's operating mechanism, mobility functions, and intended commercial use before completing the questionnaire.

To minimize interviewer bias, all enumerators received identical training and followed the same presentation script throughout the survey. The same visual materials, including photographs, the demonstration video, and technical specifications, were presented to every respondent in the same sequence to ensure consistency in the information provided. Respondents were informed that the HSEC remained a prototype under development and that their responses would not influence future purchasing opportunities. Participation was voluntary and anonymous.

3.5 Procedures to Reduce Response Bias

Several procedural measures were implemented to reduce potential response bias. First, all respondents received identical visual materials and standardized verbal explanations to ensure consistent understanding of the prototype. Second, enumerators followed a predetermined interview protocol and were instructed not to provide additional opinions or promotional statements regarding the HSEC. Third, respondents were explicitly informed that the technology was still under development and that both advantages and current limitations should be considered when evaluating purchase intention. Finally, anonymity and confidentiality were emphasized to reduce social desirability bias.

3.6 Measurement of Variables

All constructs in this study were measured using multi-item scales adapted from prior studies on green technology adoption and EV research. The measurement model includes six latent constructs: environmental concern, performance expectancy, operational cost, price, attitude toward the HSEC, and intention to purchase. Each construct was operationalized using reflective indicators measured on a five-point Likert scale, ranging from 1 ("strongly disagree") to 5 ("strongly agree"). The use of a Likert scale allows for capturing respondents' perceptions, attitudes, and behavioral intentions in a structured and quantifiable manner.

Environmental concern reflects respondents' awareness of environmental issues and their support for sustainable solutions. Performance expectancy measures perceived usefulness and expected functional benefits of the HSEC. Operational cost captures perceived cost efficiency related to energy consumption and maintenance. Price reflects perceived affordability and initial investment considerations. Attitude represents respondents' overall evaluation of the HSEC, while intention to purchase measures their willingness to adopt the technology in the future.

3.7 Data Analysis Technique

Data analysis was performed using PLS-SEM implemented in SmartPLS. The analytical procedure consisted of two sequential phases. The first phase focused on assessing the measurement model by examining indicator reliability via outer loadings, internal consistency via composite reliability, convergent validity via the average

variance extracted (AVE), and discriminant validity via both the Fornell–Larcker criterion and the heterotrait–monotrait (HTMT) ratio. The second phase involved evaluating the structural model by estimating path coefficients, coefficients of determination (R^2), effect sizes (f^2), predictive relevance, and the statistical significance of the proposed relationships using a bootstrapping procedure. PLS-SEM was selected because it is well suited to prediction-oriented research, accommodates models involving multiple latent constructs, and provides reliable parameter estimates even when survey data deviate from multivariate normality.

4. Results

4.1 Respondent Characteristics

A total of 300 valid responses were included in the final analysis. The respondents consisted of micro and small business operators engaged in mobile and semi-mobile commercial activities across the Greater Jakarta metropolitan area (Jakarta, Bogor, Depok, Tangerang, and Bekasi). Table 3 summarizes the demographic and business characteristics of the respondents.

Table 3. Respondent characteristics ($N = 300$)

Variable	Category	Frequency	Percentage (%)
Gender	Male	255	85.0
	Female	45	15.0
Age	<26 years	70	23.3
	26–35 years	89	29.7
	36–45 years	90	30.0
	46–55 years	36	12.0
	>56 years	15	5.0
	Senior High School or equivalent	289	96.3
Education	Diploma	6	2.0
	Bachelor's Degree	5	1.7
	<IDR 5 million	196	65.3
Monthly Income	IDR 5–10 million	94	31.3
	>IDR 10 million	10	3.3
Residence	Tangerang	123	41.0
	Jakarta	63	21.0
	Bogor	55	18.3
	Bekasi	39	13.0
	Depok	20	6.7
	Food vendors	167	55.7
Business Sector	Beverage vendors	66	22.0
	Retail/Grocery	2	0.7
	Other MSMEs*	65	21.6

Note: MSMEs = micro, small, and medium enterprises.

The sample was predominantly male (85.0%), reflecting the demographic composition of mobile street vendors and informal transport-dependent businesses in Indonesia, while 15.0% were female. Regarding age, respondents were relatively mature and economically active, with the largest groups aged 36–45 years (30.0%) and 26–35 years (29.7%), followed by those under 26 years (23.3%), 46–55 years (12.0%), and over 56 years (5.0%).

The respondents generally possessed a modest educational background. Most had completed senior high school or equivalent (96.3%), while only a small proportion held a diploma (2.0%) or bachelor's (1.7%) degrees. This profile is consistent with the educational characteristics commonly observed among informal MSMEs operating in urban Indonesia.

In terms of economic conditions, approximately 65.3% of respondents reported monthly incomes below IDR 5 million, whereas 31.3% earned between IDR 5–10 million, and only 3.3% reported monthly incomes exceeding IDR 10 million. These figures indicate that the surveyed MSMEs generally operate within resource-constrained environments where investment decisions are highly sensitive to operational efficiency and long-term economic benefits.

Geographically, the respondents were distributed across the Greater Jakarta metropolitan area, with the largest proportion located in Tangerang (41.0%), followed by Jakarta (21.0%), Bogor (18.3%), Bekasi (13.0%), and Depok (6.7%). The respondents represented diverse business sectors, including street food vendors, beverage sellers, traditional market traders, mobile retailers, and other informal businesses requiring daily mobility for commercial operations.

The respondents represented a diverse range of MSME business sectors. As shown in Table 3, the majority operated in the food vending sector (55.7%), followed by beverage vendors (22.0%). A smaller proportion

comprised retail and grocery businesses (0.7%), while 21.6% comprised other informal MSMEs, including mobile snack vendors, household product sellers, and various small-scale trading activities. This distribution reflects the dominant composition of transport-dependent informal enterprises operating in urban Indonesia, where food and beverage businesses constitute a substantial proportion of mobile commercial activities.

The predominance of food and beverage vendors is particularly relevant to this study's objectives because these businesses require frequent daily mobility to transport products, serve customers, and conduct business across multiple urban locations. Consequently, transportation functions not merely as a means of travel but as an essential operational resource supporting business continuity and income generation. This respondent profile therefore provides an appropriate empirical context for investigating behavioral intention toward the adoption of the HSEC among urban MSMEs.

4.2 Measurement Model Assessment

4.2.1 Indicator reliability

Figure 3 presents the measurement model evaluation for the proposed behavioral framework of purchase intention toward the HSEC. The model specifies six reflective latent constructs: environmental concern, performance expectancy, operational cost, price, attitude toward the HSEC, and intention to purchase. Conceptually, the model positions attitude as the central psychological mechanism through which environmental, technological, and economic perceptions are translated into purchase intention. This structure is appropriate for examining market readiness because adoption of an emerging renewable-energy-based mobility innovation is not determined merely by technical feasibility, but also by how potential users cognitively and affectively evaluate the innovation before forming purchase intention.

The measurement model indicates that the constructs are operationalized through multiple indicators, allowing each latent variable to capture a broader behavioral domain rather than relying on single-item measurement. This is particularly important in the context of solar-assisted micro-mobility adoption, where purchase intention may be shaped simultaneously by perceived environmental benefits, expected performance, cost efficiency, affordability, and users' overall evaluative attitude. Therefore, Figure 3 confirms that the study does not treat market acceptance as a purely economic decision, but as a multidimensional behavioral process involving sustainability awareness, utility perception, and financial rationality.

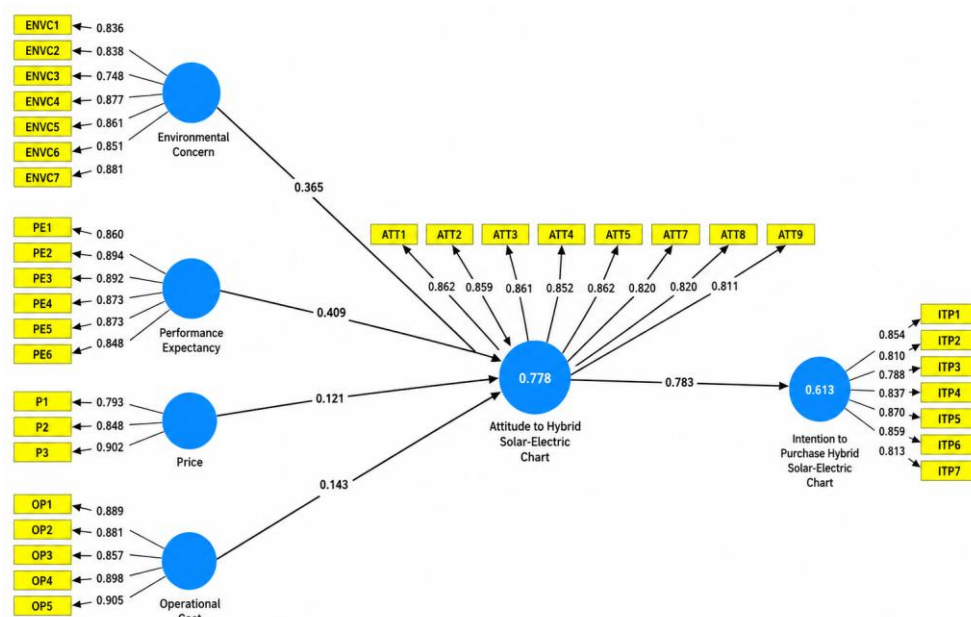


Figure 3. Measurement model evaluation

Note: ENVC = Environmental Concern, PE = Performance Expectancy, P = Price, OP = Operational Cost, ATT = Attitude to HSEC, ITP = Intention to Purchase.

Table 4 reports the outer loading values for all indicators used to measure the latent constructs in the model. The results show that all indicators have loading values above the commonly accepted threshold of 0.70, indicating satisfactory indicator reliability. The attitude construct is measured by nine indicators, with loadings ranging from 0.811 to 0.862. These values demonstrate that all attitude indicators contribute strongly and consistently to

explaining respondents' evaluative disposition toward the HSEC. The relatively narrow loading range also suggests that the attitude construct is internally stable and not dominated by a single item.

Table 4. Indicator reliability

Construct	Indicator	Outer Loading
Attitude to HSEC	ATT1	0.862
	ATT2	0.859
	ATT3	0.861
	ATT4	0.852
	ATT5	0.862
	ATT6	0.820
	ATT7	0.820
	ATT8	0.824
	ATT9	0.811
Operational Cost	OP1	0.889
	OP2	0.881
	OP3	0.857
	OP4	0.898
	OP5	0.905
Price	P1	0.793
	P2	0.848
	P3	0.902
Intention to Purchase	ITP1	0.854
	ITP2	0.810
	ITP3	0.788
	ITP4	0.837
	ITP5	0.870
	ITP6	0.859
	ITP7	0.813
Environmental Concern	ENVC1	0.836
	ENVC2	0.838
	ENVC3	0.748
	ENVC4	0.877
	ENVC5	0.861
	ENVC6	0.851
	ENVC7	0.881
Performance Expectancy	PE1	0.860
	PE2	0.894
	PE3	0.892
	PE4	0.873
	PE5	0.873
	PE6	0.848

Note: HSEC = Hybrid Solar-Electric Cart.

Environmental concern is measured by seven indicators with loadings ranging from 0.748 to 0.881. Although Environmental Concern (ENVC) 3 has the lowest loading among the environmental concern indicators, its value remains above the recommended minimum threshold, confirming its adequacy for retention. This indicates that respondents' environmental concern is captured reliably across multiple dimensions, including awareness of environmental problems and support for cleaner mobility alternatives. The high loading of ENVC7 indicates that one aspect of environmental concern may be particularly salient in shaping perceptions of solar-assisted mobility, although the construct as a whole remains balanced.

The intention to purchase construct also demonstrates strong indicator reliability, with seven indicators ranging from 0.788 to 0.870. These results indicate that purchase intention is consistently reflected across different behavioral intention items, including willingness, interest, and readiness to consider the HSEC as a future mobility solution. This finding is important because purchase intention toward prototype-based innovation may often be unstable; however, the strong loadings suggest that respondents' intention is measurable with adequate consistency.

Operational cost shows the strongest indicator reliability profile, with five indicators ranging from 0.857 to 0.905. This suggests that cost-related considerations are highly coherent in respondents' evaluation of the HSEC. In the MSME context, this is theoretically meaningful because business actors often assess mobility innovation through operational efficiency, fuel savings, maintenance implications, and daily cost reduction. The high loadings imply that operational cost is a clearly understood and behaviorally relevant construct for the target market.

Performance expectancy is measured by three indicators with loadings between 0.793 and 0.902, confirming that respondents consistently evaluate the technology based on expected usefulness and performance benefits. The

strongest loading within this construct indicates that perceived functional performance may play a particularly important role in shaping acceptance of the HSEC. For an emerging mobility prototype, this is crucial because users must be convinced that the technology can support actual business operations, not merely represent an environmentally friendly concept.

Price is measured by six indicators with loadings ranging from 0.848 to 0.894. These high values indicate that respondents interpret price-related considerations consistently. In an emerging market setting, price sensitivity is expected to be highly relevant, especially among MSMEs with limited capital capacity. The strong indicator loadings suggest that affordability, perceived price fairness, and purchase feasibility are central components of market readiness.

Table 4 confirms that the measurement indicators are statistically reliable and suitable for further validity and structural model assessment. The consistently high loadings across all constructs provide empirical support that the model has a strong measurement foundation. Substantively, the results suggest that MSME purchase intention toward the HSEC is grounded in well-defined perceptions of environmental value, performance utility, operational efficiency, affordability, and attitudinal acceptance.

4.2.2 Internal consistency reliability

Table 5 presents the composite reliability values for all latent constructs in the measurement model. The results indicate that all constructs exceed the recommended threshold of 0.70, confirming strong internal consistency reliability. Attitude toward the HSEC records the highest composite reliability value of 0.956, followed by performance expectancy at 0.951, operational cost at 0.948, environmental concern at 0.945, intention to purchase at 0.941, and price at 0.885. These findings demonstrate that the indicators used to measure each construct are highly consistent in representing their respective latent variables.

Table 5. Composite reliability

Construct	Composite Reliability
Attitude to HSEC	0.956
Environmental Concern	0.945
Intention to Purchase HSEC	0.941
Operational Cost	0.948
Performance Expectancy	0.951
Price	0.885

Note: HSEC = Hybrid Solar-Electric Cart.

The high reliability of attitude is particularly important because attitude functions as the central mediating construct in the model. A composite reliability value of 0.956 suggests that respondents' evaluative perception of the HSEC is measured with strong consistency. This strengthens the analytical foundation for examining whether attitude can translate environmental, technological, and economic perceptions into purchase intention.

Similarly, the high reliability values for performance expectancy and operational cost indicate that respondents consistently understood the functional and economic dimensions of the technology. This is significant in the MSME context, where adoption decisions are typically shaped by practical usefulness and cost efficiency. The reliability of environmental concern also confirms that sustainability-related perceptions are measured coherently, while the reliability of intention to purchase shows that respondents' behavioral readiness toward the HSEC is captured with adequate precision.

Although price has the lowest composite reliability value among the constructs, its value of 0.885 remains well above the acceptable threshold. This indicates that price-related indicators are still reliable and suitable for further analysis. Table 5 confirms that the measurement model has strong internal consistency, supporting the robustness of subsequent validity and structural model evaluation.

4.2.3 Convergent validity

Table 6 reports the AVE values for the constructs. All AVE values exceed the minimum threshold of 0.50, indicating satisfactory convergent validity. Operational cost has the highest AVE value of 0.786, followed by performance expectancy at 0.763, price at 0.721, environmental concern at 0.710, attitude toward the HSEC at 0.708, and intention to purchase at 0.695. These results indicate that each construct explains more than half of the variance of its indicators.

The high AVE value for operational cost suggests that the indicators strongly converge in capturing respondents' perception of cost efficiency. This finding is theoretically meaningful because MSMEs tend to evaluate mobility innovation through its ability to reduce operating expenses and support business sustainability. The AVE value for performance expectancy also confirms that expected usefulness and functional benefits form a coherent construct in the adoption model.

Table 6. Average variance extracted (AVE)

Construct	AVE
Attitude to HSEC	0.708
Environmental Concern	0.710
Intention to Purchase HSEC	0.695
Operational Cost	0.786
Performance Expectancy	0.763
Price	0.721

Note: HSEC = Hybrid Solar-Electric Cart.

The AVE value for price indicates that respondents consistently associate price with affordability and purchase feasibility. This is important because renewable-energy-based mobility innovation may be perceived as desirable but difficult to adopt if the acquisition cost is considered too high. Meanwhile, the AVE values for environmental concern, attitude, and intention to purchase confirm that the psychological and behavioral dimensions of the model are measured with acceptable convergence.

Table 6 provides evidence that the constructs possess adequate convergent validity. This means that the indicators within each construct share sufficient common variance and accurately represent the intended theoretical concepts. Together with the composite reliability results in Table 6, these findings confirm that the measurement model is statistically robust and appropriate for evaluating the structural relationships among environmental concern, performance expectancy, operational cost, price, attitude, and purchase intention toward the HSEC.

4.2.4 Discriminant validity

Table 7 presents the Fornell–Larcker criterion used to assess discriminant validity among the latent constructs. The diagonal values represent the square root of AVE for each construct, while the off-diagonal values represent correlations between constructs. The results show that each diagonal value is higher than the correlations with other constructs: attitude = 0.841, environmental concern = 0.843, intention to purchase = 0.834, operational cost = 0.886, performance expectancy = 0.874, and price = 0.849. This confirms that each construct shares more variance with its own indicators than with other constructs in the model.

Table 7. Fornell-Larcker criterion

Construct	ATT	ENVC	ITP	OP	PE	P
ATT	0.841	-	-	-	-	-
ENVC	0.763	0.843	-	-	-	-
ITP	0.783	0.678	0.834	-	-	-
OP	0.671	0.589	0.633	0.886	-	-
PE	0.788	0.598	0.747	0.603	0.874	-
P	0.666	0.579	0.578	0.550	0.624	0.849

Note: ATT = Attitude to Hybrid Solar-Electric Cart (HSEC); ENVC = Environmental Concern; ITP = Intention to Purchase HSEC; OP = Operational Cost; PE = Performance Expectancy; P = Price; “-” means redundant symmetric matrix entries that are not reported.

Substantively, these results indicate that the six constructs capture distinct dimensions of MSME market readiness toward the HSEC. Attitude is empirically distinguishable from environmental concern, performance expectancy, operational cost, price, and purchase intention. This is important because the model assumes that attitude is not merely another form of intention, but a separate psychological evaluation that precedes purchase intention.

The relatively high correlation between attitude and performance expectancy indicates that expected usefulness is closely associated with positive evaluation of the HSEC. However, because the square root of AVE for both constructs remains higher than their correlation, the two constructs remain empirically distinct. This supports the argument that users may recognize the functional benefits of the technology, but such recognition still needs to be transformed into a favorable attitude before purchase intention emerges.

Thus, Table 7 demonstrates acceptable discriminant validity and supports the conceptual structure of the model. The constructs are sufficiently related to form an integrated behavioral framework, yet sufficiently distinct to represent different theoretical dimensions of adoption behavior.

Table 8 reports the Heterotrait–Monotrait Ratio values as a more rigorous assessment of discriminant validity. All HTMT values are below the conservative threshold of 0.90, indicating that discriminant validity is achieved across all construct pairs. The highest HTMT value is found between performance expectancy and attitude at 0.835, followed by attitude and intention to purchase at 0.832, and environmental concern and attitude at 0.810. Although these relationships are relatively strong, they remain below the critical threshold, confirming that the constructs are not redundant.

Table 8. Heterotrait-Monotrait ratio (HTMT)

Construct	ATT	ENVC	ITP	OP	PE	P
ATT	-	-	-	-	-	-
ENVC	0.810	-	-	-	-	-
ITP	0.832	0.728	-	-	-	-
OP	0.713	0.629	0.680	-	-	-
PE	0.835	0.637	0.798	0.644	-	-
P	0.762	0.670	0.665	0.634	0.715	-

Note: ATT = Attitude to Hybrid Solar-Electric Cart (HSEC); ENVC = Environmental Concern; ITP = Intention to Purchase HSEC; OP = Operational Cost; PE = Performance Expectancy; P = Price; “-” means redundant symmetric matrix entries that are not reported.

The HTMT results strengthen the evidence obtained from the Fornell–Larcker criterion. In particular, the strong but acceptable HTMT value between attitude and intention to purchase suggests that respondents’ favorable evaluation of the HSEC is closely associated with their willingness to buy it, but the two constructs are conceptually and statistically different. This distinction is essential because the study aims to examine attitude as a mediating mechanism rather than treating it as identical to purchase intention.

The HTMT value between performance expectancy and attitude also provides an important behavioral insight. It suggests that perceived functional performance is one of the strongest sources of positive attitude formation. For MSME users, this implies that the HSEC must be perceived as reliable, useful, and operationally relevant before it can generate stronger market acceptance. Meanwhile, lower HTMT values involving operational cost and price indicate that financial considerations are related to other adoption dimensions but do not overlap excessively with them. Thus, Table 8 confirms that the measurement model has adequate discriminant validity under a stricter criterion. This gives stronger methodological confidence that subsequent structural relationships are not distorted by construct redundancy or measurement overlap.

4.3 Structural Model Assessment

4.3.1 Collinearity and common method bias assessment

Before evaluating the structural relationships, collinearity was assessed using the variance inflation factor (VIF). Table 9 shows that all inner VIF values were below the recommended threshold of 3.3, indicating that multicollinearity was not a serious concern in the structural model. Environmental concern had a VIF value of 1.910, operational cost 1.864, performance expectancy 2.078, and price 1.899 in predicting attitude toward the HSEC. Meanwhile, attitude had a VIF value of 1.000 in predicting purchase intention.

Table 9. Inner variance inflation factor (VIF) values

Relationship	VIF
Environmental Concern → Attitude	1.910
Operational Cost → Attitude	1.864
Performance Expectancy → Attitude	2.078
Price → Attitude	1.899
Attitude → Purchase Intention	1.000

These results indicate that the predictor constructs were sufficiently distinct and that the estimated structural paths were not distorted by excessive collinearity. Although full collinearity VIF values were below 3.3, this diagnostic approach primarily addresses structural bias and may not fully eliminate concerns regarding social desirability or consistency-related response tendencies commonly associated with self-reported cross-sectional surveys.

4.3.2 Coefficient of determination (R^2)

Table 10 presents the coefficient of determination values for the endogenous constructs in the structural model. The R^2 value for attitude toward the HSEC is 0.777, with an adjusted R^2 of 0.773. This indicates that environmental concern, operational cost, performance expectancy, and price collectively explain 77.7% of the variance in attitude. This represents substantial explanatory power and suggests that the model captures the major determinants shaping respondents’ evaluation of the HSEC.

This finding is theoretically important because it shows that attitude toward renewable-energy-based micro-mobility is not formed randomly or merely through general interest in innovation. Instead, attitude is strongly shaped by a combination of environmental awareness, perceived performance, operational cost efficiency, and price perception. Among MSME actors, this implies that a positive attitude emerges when the technology is perceived as environmentally meaningful, practically useful, economically efficient, and financially reasonable.

Table 10. R^2 and adjusted R^2 value

Construct	R^2	R^2 Adjusted
Attitude to HSEC	0.777	0.773
Intention to Purchase HSEC	0.613	0.612

Note: HSEC = Hybrid Solar-Electric Cart.

The R^2 value for intention to purchase is 0.613, with an adjusted R^2 of 0.612. This means that attitude explains 61.3% of the variance in purchase intention. The value indicates strong predictive relevance, especially for behavioral research involving emerging technologies and prototype-based innovations. It demonstrates that attitude is a powerful psychological predictor of market acceptance for the HSEC.

Substantively, this result suggests that commercialization efforts should not focus only on presenting the technical specifications of the cart. Instead, innovators and policymakers need to build favorable attitudes among potential users by communicating the technology's operational usefulness, cost-saving potential, environmental contribution, and affordability. When these perceptions are successfully integrated into a positive attitude, purchase intention becomes substantially stronger. Thus, Table 10 confirms that the proposed model has high explanatory capacity. The model explains both the formation of attitude and the development of purchase intention with strong predictive strength, supporting its suitability for analyzing market readiness of solar-assisted micro-mobility among urban MSMEs in Indonesia.

4.3.3 Effect size (f^2)

Table 11 presents the effect size values used to assess the substantive contribution of each exogenous construct to the endogenous constructs. The results show that attitude has a very large effect on intention to purchase, with an f^2 value of 1.587. This indicates that attitude is not only statistically relevant, but also substantively dominant in explaining purchase intention toward the HSEC. In behavioral terms, respondents' willingness to purchase the cart is strongly determined by whether they develop a favorable overall evaluation of the technology.

Table 11. Effect size (f^2) values

Construct	ATT	ENVC	ITP	OP	PE	P
ATT	-	-	1.587	-	-	-
ENVC	0.312	-	-	-	-	-
ITP	-	-	-	-	-	-
OP	0.049	-	-	-	-	-
PE	0.360	-	-	-	-	-
P	0.034	-	-	-	-	-

Note: ATT = Attitude to Hybrid Solar-Electric Cart (HSEC); ENVC = Environmental Concern; ITP = Intention to Purchase HSEC; OP = Operational Cost; PE = Performance Expectancy; P = Price; “-” denotes non-existent structural paths with no calculated effect size values.

For the predictors of attitude, performance expectancy has the largest effect size, with an f^2 value of 0.360. This indicates a medium-to-large effect and suggests that perceived usefulness and expected performance are the most influential factors in shaping positive attitudes. This finding is highly meaningful in the MSME context because business users are likely to evaluate the HSEC primarily through its practical contribution to daily operations, mobility efficiency, and business productivity.

Environmental concern also shows a meaningful effect on attitude, with an f^2 value of 0.312. This indicates a medium effect and confirms that sustainability awareness remains an important driver of attitude formation. However, its effect is slightly lower than performance expectancy, suggesting that environmental motivation alone may not be sufficient to create strong market acceptance unless the technology is also perceived as functionally useful.

Operational cost has a small effect on attitude, with an f^2 value of 0.049. This suggests that cost-efficiency considerations contribute to attitude formation, but their independent explanatory power is relatively limited compared with performance expectancy and environmental concern. Price also has a small effect, with an f^2 value of 0.034. Although price matters, its effect size indicates that affordability alone is not the primary determinant of attitude when respondents evaluate this type of innovation.

Table 11 reveals an important strategic insight: the adoption of the HSEC is primarily attitude-driven, and attitude is shaped most strongly by perceived performance and environmental concern. Therefore, market introduction strategies should emphasize the cart's operational usefulness and sustainability value, while still addressing cost and price concerns as supporting factors.

4.3.4 Predictive relevance assessment (Q^2)

Predictive relevance of the proposed structural model was further evaluated using the Stone–Geisser Q^2 statistic obtained through the blindfolding procedure. According to Hair et al. (2017), Q^2 values greater than zero indicate

that the structural model possesses predictive relevance for the endogenous constructs.

The blindfolding results demonstrate positive Q^2 values for both endogenous constructs. Attitude toward the HSEC obtained a Q^2 value of 0.544, while intention to purchase achieved a Q^2 value of 0.417 (Table 12). Since both values substantially exceed zero, the proposed model demonstrates satisfactory predictive relevance.

Table 12. Predictive relevance (Q^2)

Endogenous Construct	SSO	SSE	Q^2	Interpretation
Attitude Toward the HSEC	2349	1072.262	0.544	Strong predictive relevance
Intention to Purchase the HSEC	1827	1064.803	0.417	Strong predictive relevance

Note: HSEC = Hybrid Solar-Electric Car; SSO = sum of squares of observations; SSE = sum of squares of prediction errors.

These findings indicate that the proposed behavioral model possesses not only substantial explanatory power, as evidenced by the R^2 values, but also meaningful predictive capability for the endogenous constructs. Specifically, the model effectively predicts both MSMEs' attitudes toward the HSEC and their subsequent purchase intention. The relatively high Q^2 values suggest that integrating environmental concern, performance expectancy, operational cost, and price provides a robust basis for predicting behavioral evaluations of renewable-energy-assisted micro-mobility among MSMEs.

The positive Q^2 values strengthen the robustness of the structural model and provide additional support for its suitability as a prediction-oriented framework in accordance with the objectives of PLS-SEM.

4.3.5 Out-of-sample predictive assessment (PLSpredict)

To further evaluate the model's predictive capability beyond explanatory performance, an out-of-sample prediction assessment was conducted using the PLSpredict procedure. Following the recommendations of Hair et al. (2017), the predictive performance of the PLS-SEM model was compared with that of a naïve linear regression benchmark (LM) using the Root Mean Squared Error (RMSE) as the primary prediction error metric.

The comparison indicates that the PLS-SEM model generally outperformed the linear benchmark. Specifically, 14 of the 16 measurement indicators (87.5%) exhibited lower RMSE values under the PLS model than under the linear model (Table 13). For the remaining two indicators, the prediction errors were very similar between the two models, suggesting only marginal differences in predictive accuracy.

Furthermore, all indicators produced positive Q^2_{predict} values, providing additional evidence that the model possesses meaningful out-of-sample predictive capability. Collectively, these findings indicate that the proposed structural model not only explains the observed relationships among the constructs but also demonstrates satisfactory predictive performance when applied to new observations.

Table 13. Summary of PLSpredict results

Construct	Number of Indicators	PLS RMSE < LM RMSE	Interpretation
Attitude	9	9	High predictive power
Purchase Intention	7	5	High predictive power
Overall	16	14 (87.5%)	High predictive power

According to the predictive assessment criteria proposed by Hair et al. (2017), because the majority of indicators exhibit lower prediction errors in the PLS model than in the linear benchmark, the model demonstrates high predictive power. These results complement the R^2 and Stone–Geisser Q^2 statistics and further strengthen the robustness of the proposed behavioral framework for explaining MSMEs' adoption intention toward the HSEC.

4.3.6 Model fit assessment

Table 14 reports the model fit indices for both the saturated and estimated models. The SRMR value is 0.054 for the saturated model and 0.063 for the estimated model. Both values are below the commonly accepted cut-off of 0.08, indicating acceptable model fit. This suggests that the discrepancy between the observed correlation matrix and the model-implied correlation matrix is relatively low.

Table 14. Model fit

Fit Index	Saturated Model	Estimated Model
SRMR	0.054	0.063
d_{ULS}	2.079	2.834
d_{G}	1.199	1.238
Chi-Square	1,769.339	1,809.601
Normed Fit Index (NFI)	0.817	0.812

The d_{ULS} and d_G values provide additional information about model discrepancy. The estimated model shows d_{ULS} of 2.834 and d_G of 1.238. These values should be interpreted cautiously in PLS-SEM because model fit assessment in variance-based SEM is primarily complementary rather than definitive. Nevertheless, the reported values do not contradict the earlier evidence of acceptable measurement quality and explanatory power.

The NFI value is 0.817 for the saturated model and 0.812 for the estimated model. Although this value is below the conventional 0.90 benchmark often used in covariance-based SEM, it remains acceptable in exploratory or prediction-oriented PLS-SEM studies, particularly when the model involves behavioral constructs and emerging technology adoption. More importantly, the SRMR values indicate that the model has an acceptable level of approximate fit.

Substantively, Table 14 supports the adequacy of the proposed theoretical structure. The model is sufficiently aligned with the empirical data to justify further interpretation of the structural paths. This means that the relationships among environmental concern, performance expectancy, operational cost, price, attitude, and purchase intention can be interpreted with reasonable confidence. The acceptable fit also reinforces the argument that the proposed behavioral framework is suitable for analyzing the market readiness of renewable-energy-based micro-mobility innovation in urban Indonesia.

4.4 Hypothesis Testing

The structural model was evaluated using path coefficients, t -statistics, p -values, and bootstrapped confidence intervals. Bootstrapping results indicate that all direct relationships were positive and statistically significant. Attitude toward the HSEC was positively associated with intention to purchase ($\beta = 0.783$, $t = 31.724$, $p < 0.001$), indicating that respondents with a more favorable evaluation of the HSEC were more likely to report purchase intention (see Table 15 and Figure 4).

Among the antecedents of attitude, performance expectancy showed the strongest positive association ($\beta = 0.409$, $t = 8.083$, $p < 0.001$), followed by environmental concern ($\beta = 0.365$, $t = 8.439$, $p < 0.001$). Operational cost ($\beta = 0.143$, $t = 3.312$, $p = 0.001$) and price ($\beta = 0.121$, $t = 2.598$, $p = 0.009$) were also positively associated with attitude, although their effect sizes were smaller. These findings suggest that MSME respondents evaluated the HSEC primarily through expected functional usefulness and environmental value, while cost and price considerations played supporting roles.

Table 15. Direct effects with confidence intervals

Hypothesis	Relationship	β	t -Value	p -Value	95% Confidence Interval
H1	Environmental Concern $\rightarrow$ Attitude	0.365	8.439	<0.001	0.277–0.447
H2	Performance Expectancy $\rightarrow$ Attitude	0.409	8.083	<0.001	0.310–0.508
H3	Operational Cost $\rightarrow$ Attitude	0.143	3.312	0.001	0.060–0.230
H4	Price $\rightarrow$ Attitude	0.121	2.598	0.009	0.026–0.210
H5	Attitude $\rightarrow$ Purchase Intention	0.783	31.724	<0.001	0.730–0.829

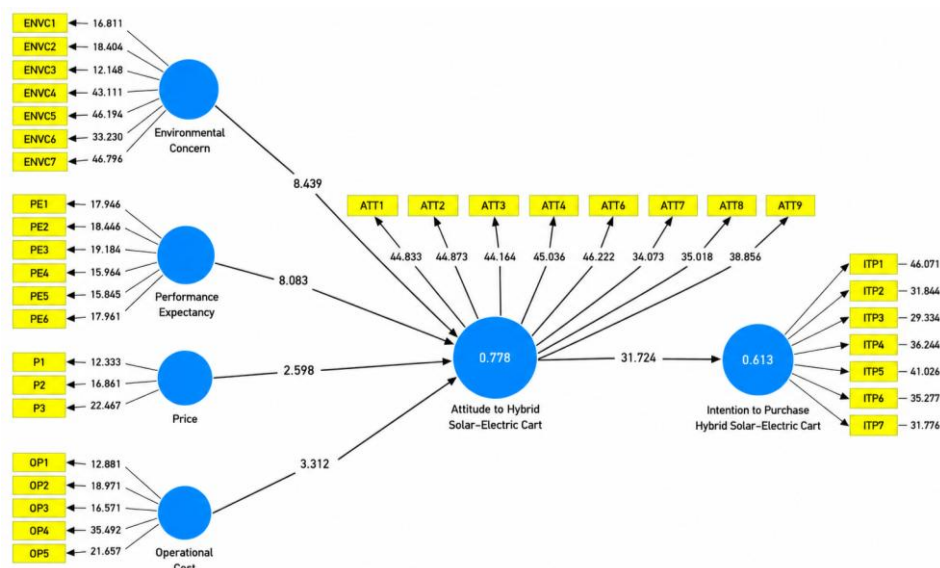


Figure 4. Structural model evaluation

Note: ENVC = Environmental Concern, PE = Performance Expectancy, P = Price, OP = Operational Cost, ATT = Attitude to HSEC, ITP = Intention to Purchase.

Performance expectancy has the strongest direct effect on attitude, with a coefficient of 0.409, a *t*-statistic of 8.083, and a *p*-value of < 0.001. This result indicates that perceived performance is the most important predictor of attitude among the four antecedent variables. In practical terms, potential users are more likely to develop a positive attitude when they believe the HSEC can improve mobility efficiency, support operational needs, and deliver reliable performance. This finding highlights the importance of demonstrating functional capability during product testing, promotion, and commercialization.

Environmental concern also has a strong and significant effect on attitude, with a coefficient of 0.365, a *t*-statistic of 8.439, and a *p*-value of < 0.001. This shows that respondents with greater environmental concern tend to evaluate the HSEC more positively. The result confirms that sustainability value is relevant to adoption behavior, particularly because the cart integrates solar energy and electric mobility. However, because the coefficient is slightly lower than performance expectancy, the finding suggests that environmental benefits must be supported by practical utility to generate stronger market acceptance.

Operational cost has a positive and significant effect on attitude, with a coefficient of 0.143, a *t*-statistic of 3.312, and a *p*-value of 0.001. This indicates that perceptions of lower operating costs contribute to a more favorable attitude toward the cart. For MSMEs, this result is important because mobility technologies are often evaluated based on their ability to reduce daily business expenses. Although the effect is smaller than performance expectancy and environmental concern, it remains statistically meaningful and indicates that cost-efficiency communication should be included in commercialization strategies.

Price also has a positive and significant effect on attitude, with a coefficient of 0.121, a *t*-statistic of 2.598, and a *p*-value of 0.009. This result suggests that perceived price fairness and affordability influence respondents' attitudes toward the HSEC. However, the relatively smaller coefficient indicates that price is not the dominant driver of attitude. This may imply that respondents are willing to consider the technology if they perceive strong performance, environmental benefits, and operational cost advantages, even if price remains an important consideration.

Table 15 demonstrates that all hypothesized direct paths are positive and statistically significant. The findings provide strong empirical support for the proposed model. More importantly, the pattern of coefficients reveals a clear hierarchy of adoption drivers: attitude is the strongest determinant of purchase intention, while performance expectancy and environmental concern are the strongest antecedents of attitude, followed by operational cost and price. This suggests that the market readiness of the HSEC depends on a combination of functional credibility, sustainability value, economic rationality, and favorable user evaluation.

4.5 Mediation Analysis

The mediation analysis examined the indirect associations of environmental concern, performance expectancy, operational cost, and price with purchase intention through attitude. The results show that all indirect effects were positive and statistically significant.

Performance expectancy had the strongest indirect association with purchase intention through attitude ($\beta = 0.320$, $t = 7.421$, $p < 0.001$), followed by environmental concern ($\beta = 0.286$, $t = 8.477$, $p < 0.001$). Operational cost ($\beta = 0.112$, $t = 3.249$, $p = 0.001$) and price ($\beta = 0.095$, $t = 2.651$, $p = 0.008$) also showed significant indirect associations through attitude (see Table 16).

Table 16. Specific indirect effects

Hypothesis	Indirect Relationship	β	<i>t</i> -Value	<i>p</i> -Value	95% Confidence Interval
H6a	Environmental Concern → Attitude → Purchase Intention	0.286	8.477	<0.001	0.217–0.350
H6b	Performance Expectancy → Attitude → Purchase Intention	0.320	7.421	<0.001	0.236–0.405
H6c	Operational Cost → Attitude → Purchase Intention	0.112	3.249	0.001	0.047–0.183
H6d	Price → Attitude → Purchase Intention	0.095	2.651	0.008	0.023–0.165

Table 16 presents the specific indirect effects of environmental concern, operational cost, performance expectancy, and price on intention to purchase through attitude toward the HSEC. The results show that all indirect effects are positive and statistically significant, indicating that attitude functions as an important mediating mechanism in the proposed model.

The strongest indirect effect is found in the relationship between performance expectancy and intention to purchase through attitude, with a coefficient of 0.320, *t*-statistic of 7.421, and *p*-value of < 0.001. This finding indicates that respondents' belief in the cart's functional usefulness does not merely influence their intention directly in a mechanical way; rather, expected performance first shapes a favorable attitude, which then strengthens

purchase intention. In the MSME context, this suggests that users must be convinced that the HSEC can support business mobility, operational reliability, and daily productivity before they develop a stronger willingness to purchase it.

Environmental concern also has a substantial indirect effect on intention to purchase through attitude, with a coefficient of 0.286, *t*-statistic of 8.477, and *p*-value of < 0.001. This result confirms that environmental awareness becomes behaviorally meaningful when it is translated into a positive evaluation of the product. In other words, concern for sustainability alone is not sufficient; it must be connected to the perception that the HSEC is a desirable and relevant solution for cleaner urban mobility.

Operational cost has a positive and significant indirect effect on intention to purchase through attitude, with a coefficient of 0.112, *t*-statistic of 3.249, and *p*-value of 0.001. This means that perceived operational savings strengthen purchase intention by improving respondents' attitude toward the technology. For MSMEs, this is highly relevant because adoption decisions are often linked to daily cost efficiency, fuel savings, and long-term operating benefits.

Price also shows a positive and significant indirect effect, with a coefficient of 0.095, *t*-statistic of 2.651, and *p*-value of 0.008. Although this is the smallest indirect effect, it remains statistically meaningful. This indicates that perceived price fairness contributes to purchase intention when it helps form a favorable attitude toward the HSEC. Therefore, price should not be treated merely as a financial barrier, but as part of the broader value perception that shapes user acceptance.

5. Discussion

This study contributes to the growing body of literature on sustainable mobility by examining the behavioral mechanisms underlying the adoption of hybrid solar-electric micro-mobility among MSMEs in emerging urban economies. Unlike conventional EV studies that focus on private consumers or policy-driven adoption, the present findings highlight a multi-dimensional evaluative process in which adoption intention is shaped through an integrated assessment of performance, sustainability, and economic feasibility.

5.1 Attitude as a Central Mechanism in Technology Adoption

The findings of this study highlight the pivotal role of attitude as a central mediating mechanism, rather than merely a direct predictor of behavioral intention. This extends the traditional assumptions of the TPB and TAM, where attitude is often treated as one of several parallel determinants. In the context of emerging mobility technologies, however, users appear to rely more heavily on attitudinal evaluation to process uncertainty and limited prior experience.

Recent empirical studies support this stronger conceptualization of attitude. For instance, Noor et al. (2025) demonstrate that attitude significantly mediates the relationship between perceived benefits and purchase intention in EV adoption, emphasizing its role as a cognitive integration mechanism. Similarly, Figueiredo & Baptista (2025) show that urban mobility adoption behavior is strongly shaped by users' evaluative judgments, particularly in environments characterized by technological transition.

The findings indicate that attitude represents the strongest predictor of purchase intention toward the HSEC. Rather than functioning merely as an affective evaluation, attitude appears to integrate environmental, technological, and economic considerations into an overall judgment regarding the feasibility of adopting the innovation. This finding is particularly relevant because respondents evaluated a prototype technology with which they had no prior ownership experience. Under such circumstances, prospective adopters rely on their overall evaluation of the technology before expressing behavioral intention.

The present findings extend previous technology adoption studies by demonstrating that attitude plays a particularly important role within MSME decision-making. Unlike household consumers, MSME operators evaluate transportation technologies not only as mobility solutions but also as productive business assets that directly influence operational continuity and income generation. Consequently, attitude reflects a broader assessment encompassing expected business benefits, operational feasibility, sustainability value, and investment viability.

This interpretation aligns with recent research showing that attitude remains a critical predictor of EV adoption in emerging economies. However, the present study suggests that within business-oriented mobility contexts, attitude represents an entrepreneurial evaluation rather than merely a consumer preference. This distinction contributes to the growing literature on sustainable mobility by extending behavioral adoption models from household transportation toward enterprise-oriented renewable mobility systems (Adzhani et al., 2025; Lazuardy et al., 2025).

5.2 Performance Expectancy and Functional Rationality

The strong influence of performance expectancy reflects a functional rationality framework, particularly relevant in MSME contexts. Unlike individual consumers, MSMEs evaluate technology primarily based on its ability to enhance operational efficiency, productivity, and reliability.

This finding is consistent with recent studies emphasizing the importance of perceived usefulness in green technology adoption. Rahmani & Zamani (2026) demonstrate that performance-related attributes significantly influence both attitude and intention toward electric mobility adoption. Likewise, Zhao et al. (2024) confirm that perceived functional benefits are among the most critical drivers of behavioral intention.

Performance expectancy emerged as the strongest antecedent of attitude, indicating that MSME operators primarily evaluate the HSEC according to its expected ability to support business activities. This result differs from many consumer-oriented EV studies, where performance is frequently assessed in terms of driving comfort, convenience, or personal mobility. In contrast, respondents in the present study viewed vehicle performance through an operational lens, emphasizing reliability, carrying capacity, travel range, and the ability to sustain daily commercial activities.

This finding reflects the operational realities of Indonesian MSMEs, particularly mobile and semi-mobile enterprises operating in dense urban environments. Street vendors, beverage sellers, traditional market traders, and mobile retailers depend heavily on transportation to reach customers, transport merchandise, and maintain business continuity. Consequently, any uncertainty regarding vehicle reliability or operational capability may directly affect daily income. Under these conditions, technological performance becomes an essential business consideration rather than simply a desirable product attribute.

The finding is also consistent with broader evidence that Indonesia's electric mobility transition remains concentrated in personal transportation, while commercial micro-mobility applications are still at an early stage. This highlights the importance of demonstrating practical operational performance when introducing renewable-energy mobility innovations for MSMEs. Product demonstrations, pilot projects, and field trials may therefore be more effective than conventional promotional campaigns because they reduce uncertainty surrounding real-world business performance.

However, the present study extends these findings by showing that performance expectancy is not only influential but also foundational. In the absence of proven functionality, other factors such as environmental concern and price become less relevant. This aligns with task-technology fit theory, which posits that adoption occurs only when technology capabilities match user needs.

5.3 Environmental Concern as Conditional Driver

Environmental concern plays a significant but conditional role in shaping adoption behavior. While users increasingly recognize the importance of sustainability, their willingness to adopt green technologies depends on the alignment between environmental benefits and practical utility.

Recent research supports this interpretation. Tolani et al. (2025) find that environmental awareness has a significant influence on the adoption of sustainable mobility, particularly in developing economies. Similarly, Rahman et al. (2025) demonstrate that environmental concern positively affects EV adoption, but its impact is mediated by perceived usefulness and economic considerations.

Environmental concern also showed a significant positive association with attitude, indicating that sustainability considerations contribute to favorable evaluations of the HSEC. However, its influence was slightly weaker than performance expectancy, suggesting that environmental values alone are insufficient to motivate adoption among resource-constrained MSMEs.

This finding reflects the practical realities faced by many Indonesian micro-enterprises. Although awareness of environmental sustainability continues to increase, daily business survival remains the dominant priority. Entrepreneurs operating within the informal economy frequently face uncertain income, limited access to formal credit, fluctuating fuel prices, and highly competitive market conditions. Consequently, environmentally friendly technologies are more likely to be accepted when environmental benefits are accompanied by tangible operational and economic advantages.

Rather than contradicting previous research, these findings suggest that environmental concern functions as a conditional driver of technology adoption. Sustainability becomes behaviorally meaningful when entrepreneurs perceive that environmental responsibility is compatible with business productivity and long-term economic resilience. Thus, environmental communication alone may have limited effectiveness unless accompanied by clear evidence regarding operational and financial benefits.

5.4 Economic Considerations Within a Value-Based Framework

Economic factors, including operational cost and price, remain important but are not the primary drivers of

adoption. Instead, they function as components of a broader value-based evaluation, where users assess cost relative to expected benefits. This perspective is supported by recent literature. Tolani et al. (2025) highlight that cost remains a barrier, particularly due to high initial investment requirements. However, Li et al. (2025) show that perceived value, rather than absolute cost, plays a more decisive role in shaping adoption intention.

Operational cost and price were positively associated with attitude, although their standardized coefficients were smaller than those of performance expectancy and environmental concern. These findings indicate that economic considerations remain important but are evaluated within a broader value-based framework rather than as isolated financial barriers.

For Indonesian MSMEs, transportation expenditure represents a recurring component of daily operating costs. Reductions in fuel consumption and maintenance expenses directly improve operating margins and may enhance long-term business sustainability. Nevertheless, respondents did not appear to evaluate the HSEC solely according to its purchase price. Instead, they considered whether the technology could generate sufficient operational benefits to justify the required investment.

This interpretation reflects entrepreneurial decision-making, where investment appraisal extends beyond initial acquisition cost toward expected lifetime value. Therefore, policies aimed solely at reducing purchase prices may not fully address adoption barriers. Financing mechanisms that reduce investment risk, such as lease-to-own schemes, micro-credit programs, or cooperative financing, may prove more effective because they allow MSMEs to realize operational benefits while gradually recovering investment costs.

5.5 Implications for Sustainable Urban Mobility in Indonesia

Beyond its theoretical contributions, this study provides practical insights for accelerating sustainable urban mobility among MSMEs in Indonesia. Current national EV policies primarily emphasize passenger vehicles and electric motorcycles, whereas renewable-energy-assisted commercial micro-mobility remains largely unexplored. The findings suggest that commercialization strategies for technologies such as the HSEC should focus on improving business productivity while simultaneously delivering environmental benefits.

At the local government level, pilot-scale demonstration projects in traditional markets, street-vendor centers, and urban commercial districts could reduce uncertainty regarding the operational performance of renewable-energy mobility technologies. Municipal governments could collaborate with universities, MSME cooperatives, financial institutions, and private manufacturers to establish demonstration programs and shared solar-charging facilities specifically designed for small businesses.

Financial accessibility also represents an important policy consideration. Rather than relying exclusively on direct purchase subsidies, financing mechanisms tailored to MSMEs, including lease-to-own arrangements, cooperative financing, revolving micro-credit, or public-private partnership schemes, may reduce investment barriers while maintaining commercial sustainability. Such approaches are particularly relevant in emerging economies, where many informal entrepreneurs experience limited access to conventional financing.

Overall, the findings indicate that successful diffusion of renewable-energy-assisted micro-mobility requires more than technological innovation. It requires an integrated ecosystem combining appropriate technology, accessible financing, institutional support, and business-oriented commercialization strategies capable of addressing the operational realities faced by urban MSMEs. These considerations may facilitate Indonesia's broader transition toward sustainable urban mobility while simultaneously strengthening the competitiveness and resilience of the MSME sector.

5.6 Theoretical Contributions

The present study makes several theoretical contributions to the literature on sustainable mobility, renewable-energy innovation, and technology adoption.

First, this study extends the application of established behavioral adoption theories to a business-oriented renewable mobility context. Previous studies grounded in the TPB, the TAM, and the UTAUT have predominantly examined the adoption of passenger EVs by individual consumers. In those contexts, technology adoption is generally conceptualized as a personal mobility decision influenced by individual preferences, perceived usefulness, environmental values, and economic considerations. By contrast, the present study demonstrates that adoption of the HSEC among MSMEs represents a fundamentally different behavioral setting in which transportation functions as an operational business resource rather than solely as a means of personal travel. Consequently, the findings broaden the applicability of established technology adoption theories beyond consumer mobility toward enterprise-oriented renewable mobility systems operating in emerging economies.

Second, the study provides empirical evidence that established behavioral constructs operate within a different decision-making logic when mobility technologies are evaluated as productive business assets. Although environmental concern, performance expectancy, operational cost, price, attitude, and purchase intention have been widely investigated in previous EV studies, the present findings indicate that these constructs should not be

interpreted identically across different adoption contexts. MSME operators simultaneously evaluate environmental sustainability, operational productivity, investment feasibility, and long-term business continuity. This multidimensional evaluation process suggests that technology adoption among entrepreneurs involves a broader business-oriented assessment than conventional consumer-oriented adoption models generally assume.

Third, the study contributes to the literature by demonstrating the dominant role of performance expectancy within resource-constrained entrepreneurial environments. While previous studies frequently identify purchase price as one of the principal barriers to EV adoption, the present findings indicate that perceived operational capability exerts the strongest influence on attitude toward the HSEC. This suggests that for MSMEs, technological functionality and operational reliability outweigh initial acquisition cost because transportation directly affects daily commercial activities and income generation. The findings therefore support the proposition that entrepreneurial adoption decisions are driven primarily by expected business performance rather than solely by financial affordability.

Fourth, this study advances understanding of attitude as a psychological integration mechanism within prototype-based renewable-energy innovations. Rather than functioning merely as one predictor among several behavioral determinants, attitude integrates environmental, technological, and economic evaluations before purchase intention is formed. This mediating role becomes particularly important when users evaluate technologies that remain in the pre-commercialization stage and therefore have limited market experience. The findings suggest that prospective adopters first construct an overall evaluative judgment regarding the innovation before expressing behavioral intention, highlighting the importance of attitude in the early commercialization of sustainable technologies.

Finally, the study contributes to the commercialization literature on renewable-energy innovation by examining purchase intention toward a technology that has not yet entered large-scale commercial deployment. Most previous adoption studies investigate technologies that are already available in the marketplace and whose characteristics are relatively familiar to consumers. In contrast, the present research evaluates behavioral intention toward an emerging prototype specifically developed for urban MSMEs. By focusing on the pre-commercialization stage, this study provides empirical evidence regarding the behavioral factors associated with early market acceptance and offers a conceptual bridge between technological innovation and commercialization strategy. This perspective contributes to a broader understanding of how renewable-energy-assisted mobility innovations may successfully transition from technological prototypes to commercially viable sustainable transportation solutions.

Collectively, these theoretical contributions suggest that renewable-energy-assisted micro-mobility should not be viewed simply as a smaller version of conventional EVs. Instead, it represents a distinct category of sustainable transportation in which environmental objectives, technological performance, operational efficiency, and entrepreneurial decision-making converge. Accordingly, future technology adoption research should increasingly distinguish between consumer-oriented mobility and business-oriented mobility, as the behavioral mechanisms underlying these two contexts may differ substantially despite employing similar theoretical constructs.

5.7 Future Research Directions

While the present study provides important insights into the behavioral determinants associated with the adoption of renewable-energy-assisted micro-mobility among MSMEs, several opportunities remain for future research.

First, future studies should investigate actual adoption behavior rather than purchase intention. The present research examined prospective behavioral intention because the HSEC remains at the prototype stage and has not yet been commercialized on a large scale. Consequently, respondents evaluated the technology based on its expected characteristics rather than direct ownership or operational experience. As commercialization progresses, longitudinal studies following MSMEs from initial evaluation through actual adoption and continued use would provide a more comprehensive understanding of technology diffusion and post-adoption behavior.

Second, future research should expand the geographical scope beyond the Greater Jakarta metropolitan area. Indonesia exhibits substantial regional diversity in terms of infrastructure availability, transportation systems, business characteristics, and economic development. MSMEs operating in secondary cities or rural regions may face different mobility requirements and institutional constraints that influence renewable-energy technology adoption. Comparative studies across provinces or between urban and rural areas would therefore enhance the external validity of the present findings.

Third, future research should incorporate additional constructs that are particularly relevant to business-oriented mobility adoption. While the present study focuses on environmental concern, performance expectancy, operational cost, price, attitude, and purchase intention, entrepreneurial adoption decisions may also be influenced by factors such as business model compatibility, maintenance autonomy, perceived operational risk, financing accessibility, government incentives, charging infrastructure availability, and organizational readiness. Integrating these variables into existing behavioral adoption models may provide a more comprehensive explanation of renewable-energy technology adoption among MSMEs.

Fourth, future studies may adopt multi-group or comparative research designs to examine whether adoption behavior differs across various types of MSMEs. Food vendors, beverage sellers, mobile retailers, logistics providers, and service-oriented enterprises operate under different business conditions, carrying capacities, travel distances, and daily mobility requirements. Understanding these sector-specific differences would help identify market segments in which renewable-energy-assisted micro-mobility offers the greatest practical and economic value.

Fifth, future investigations should consider evaluating the long-term economic and environmental performance of the HSEC under real operating conditions. Combining behavioral research with engineering performance assessments and life-cycle economic analysis would provide stronger evidence regarding the technology's commercial feasibility, operational reliability, maintenance requirements, and environmental benefits. Such interdisciplinary approaches would facilitate a more comprehensive evaluation of renewable-energy-assisted mobility systems and support evidence-based commercialization strategies.

Finally, comparative international research would further strengthen understanding of renewable-energy-assisted micro-mobility adoption in emerging economies. Many developing countries share similar challenges, including rapid urbanization, informal economic activities, limited charging infrastructure, and constrained access to financing for small businesses. Cross-country comparative studies would enable researchers to distinguish universal behavioral mechanisms from those shaped by local institutional, cultural, and economic conditions, thereby contributing to the broader literature on sustainable urban mobility and renewable-energy innovation.

Collectively, these future research directions highlight that renewable-energy-assisted micro-mobility remains an emerging research field with considerable theoretical and practical potential. As prototype technologies progress toward commercialization, future studies integrating behavioral, engineering, economic, and policy perspectives will be essential for advancing sustainable transportation systems that simultaneously promote environmental sustainability, technological innovation, and inclusive economic development.

5.8 Study Limitations

This study should be interpreted in light of several limitations. First, the research employed a cross-sectional survey design, which captures respondents' perceptions at a single point in time. Accordingly, although the proposed model identifies significant associations among the constructs, causal relationships cannot be inferred. Longitudinal research following respondents throughout different stages of technology adoption would provide stronger evidence regarding causal mechanisms and behavioral changes over time.

Second, the study utilized convenience sampling because a comprehensive sampling frame for mobile and semi-mobile MSMEs was unavailable. Although this approach is commonly adopted in behavioral studies involving informal business sectors, the findings should be generalized cautiously beyond the surveyed population.

Third, the respondents were drawn exclusively from the Greater Jakarta metropolitan area. Consequently, the results may not fully represent MSMEs operating in other Indonesian regions with different transportation infrastructures, economic conditions, or business environments.

Fourth, the study investigated purchase intention toward a prototype technology rather than actual adoption behavior. Although respondents received standardized photographs, technical specifications, and a demonstration video explaining the operation of the HSEC, their evaluations remained based on expected rather than experienced performance. Actual user experience following commercialization may produce different behavioral responses.

Finally, all variables were measured using self-reported questionnaires administered during a single survey period. Although several procedural measures were implemented to minimize common method bias, including standardized presentation materials, identical survey procedures, and anonymous participation, some degree of common method variance may still exist. Future studies incorporating objective operational data, longitudinal observations, or multiple data sources would further strengthen the robustness of the findings.

6. Conclusions

This study provides empirical evidence on the behavioral mechanisms underlying the adoption of solar-assisted micro-mobility as a pathway toward sustainable urban mobility in emerging economies. The results demonstrate significant indirect relationships through attitude, highlighting its central mediating role in shaping purchase intention. This indicates that users must first develop a coherent evaluative perception of the technology before forming a concrete intention to adopt it.

Among the determinants, performance expectancy emerges as the most influential driver, highlighting the centrality of functional reliability and operational usefulness in shaping adoption decisions. Environmental concern also plays a significant role, confirming that sustainability awareness contributes to positive evaluations when aligned with practical benefits. In contrast, operational cost and price, while statistically significant, function as supporting factors within a broader value-based assessment rather than as primary decision drivers.

From a sustainability perspective, the study advances the understanding of how renewable energy-based

mobility innovations can transition from technological feasibility to market acceptance. The results suggest that successful diffusion of solar-assisted micro-mobility depends on integrating performance credibility, environmental value, and economic feasibility into a unified user perception. This reinforces the importance of aligning sustainability goals with real-world operational needs, particularly in resource-constrained MSME contexts.

The study contributes to the literature by proposing a holistic behavioral adoption framework that integrates environmental, technological, and economic dimensions within a unified evaluative process.

In terms of practical implications, policymakers should prioritize demonstration-based interventions, pilot programs, and user education initiatives to strengthen perceived performance and build user confidence. Financial incentives, such as subsidies or flexible financing schemes, should be designed to complement rather than substitute functional value. For industry stakeholders, value-based communication strategies emphasizing reliability, cost efficiency, and sustainability benefits are essential to accelerate adoption among MSMEs.

This study has several limitations. First, the use of a cross-sectional design limits the ability to capture dynamic changes in user perception over time. Second, the reliance on intention-based measures may not fully reflect actual adoption behavior. Third, the study focuses on urban MSMEs in Indonesia, which may limit the generalizability of the findings to other sectors or geographical contexts.

Future research should adopt longitudinal approaches to examine the transition from intention to actual adoption and incorporate real-world usage data. Comparative studies across different regions or types of users would provide deeper insights into contextual variations in adoption behavior. Additionally, integrating policy, infrastructure, and ecosystem-level variables into the analytical framework would further enrich the understanding of sustainable mobility transitions.

Author Contributions

Conceptualization, E.P. and H.N.; methodology, E.P.; software, E.P.; validation, E.P. and I.A.S.; formal analysis, E.P.; investigation, E.P., H.N., and I.A.S.; data curation, E.P.; writing—original draft preparation, E.P.; writing—review and editing, E.P., H.N., and I.A.S.; visualization, E.P.; supervision, E.P.; project administration, E.P. All authors have read and agreed to the published version of the manuscript.

Funding

The authors gratefully acknowledge financial support provided by the Directorate of Research and Community Service (DPPM), Ministry of Higher Education, Science, and Technology of the Republic of Indonesia (Kemendikristek), through the Multiyear Research Grant Scheme (Grant No.: 131/C3/DT.05.00/PL-MULTITAHUN LANJUTAN/2026; 2001/LL4/PG/2026; 011/PKS-LP2M/UPJ/05.26).

Informed Consent Statement

Informed consent was obtained from all subjects involved in the study. All respondents participated voluntarily and were informed about the purpose of the research before data collection.

Ethical Approval

This research complied with generally accepted ethical principles for scientific investigation. As the study neither involved sensitive personal information, vulnerable participants, nor experimental interventions, formal ethical clearance from an institutional review board was not considered necessary.

Data Availability

The data presented in this study are available from the corresponding author upon reasonable request. The data are not publicly available due to privacy considerations and confidentiality agreements with respondents.

Acknowledgements

Appreciation is extended to Universitas Pembangunan Jaya for its institutional support and to all MSME participants whose valuable time and insights made this study possible.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors declare that generative artificial intelligence (AI) tools were used solely to assist in language refinement, grammar checking, and improving the clarity of the manuscript. All conceptual development, data analysis, interpretation of results, and scientific conclusions were conducted entirely by the authors. The authors take full responsibility for the content of this publication.

References

- Adzhani, I. A., Pramono, B., Santoso, M. S., Wirayat, M. Y. F., Okdinawati, L., & Belgiawan, P. F. (2025). Battery electric vehicle adoption in Indonesia: Insights from consumer preferences and stakeholder perspectives. *J. Civ. Eng. Forum*, 11(3), 334–350. <https://doi.org/10.22146/jcef.22067>.
- Ajao, Q., Prio, M. H., & Sadeeq, L. (2025). Analysis of factors influencing electric vehicle adoption in Sub-Saharan Africa using a modified UTAUT framework. *Discover Electron.*, 2(1), 1–15. <https://doi.org/10.1007/s44291-025-00043-4>.
- Balassa, B. E. & Koteczki, R. (2026). Attitudes towards sustainable mobility technologies: Insights from electric vehicle adoption. *Soc. Sci. Humanit. Open*, 13, 102379. <https://doi.org/10.1016/j.ssaho.2025.102379>.
- Buhmann, K. M., Rialp-Criado, J., & Rialp-Criado, A. (2024). Predicting consumer intention to adopt battery electric vehicles: Extending the theory of planned behavior. *Sustainability*, 16(3), 1284. <https://doi.org/10.3390/su16031284>.
- Chanda, R. C., Vafaei-Zadeh, A., Hanifah, H., Ashrafi, D. M., & Ahmed, T. (2024). Achieving a sustainable future by analyzing electric vehicle adoption in developing nations through an extended technology acceptance model. *Sustain. Futures*, 8, 100386. <https://doi.org/10.1016/j.sfr.2024.100386>.
- D’Adamo, I., Gastaldi, M., Piccioni, J., & Rosa, P. (2023). The role of automotive flexibility in supporting the diffusion of sustainable mobility initiatives: A stakeholder attitudes assessment. *Global J. Flexible Syst. Manage.*, 24(3), 459–481. <https://doi.org/10.1007/s40171-023-00349-w>.
- Egbue, O. & Long, S. (2012). Barriers to widespread adoption of electric vehicles: An analysis of consumer attitudes and perceptions. *Energy Policy*, 48, 717–729. <https://doi.org/10.1016/j.enpol.2012.06.009>.
- Figueiredo, V. & Baptista, G. (2025). Electric vehicles sustainability and adoption factors. *Urban Sci.*, 9(8), 311. <https://doi.org/10.3390/urbansci9080311>.
- Gutjar, M., Calastri, C., & Kowald, M. (2025). Electric vehicle ownership dynamics at the household level: A stated adaptation experiment on the effects of pricing and incentive policies. *Transportation*, 1–54. <https://doi.org/10.1007/s11116-025-10611-y>.
- Hackbarth, A. & Madlener, R. (2013). Consumer preferences for alternative fuel vehicles: A discrete choice analysis. *Transp. Res. Part D Transp. Environ.*, 25, 5–17. <https://doi.org/10.1016/j.trd.2013.07.002>.
- Hair, J. F., Hult, G. T. M., Ringle, C. M., & Sarstedt, M. (2017). *A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM) (2nd ed.)*. Sage Publications.
- Hardman, S., Jenn, A., Tal, G., Axsen, J., Beard, G., Daina, N., Figenbaum, E., Jakobsson, N., Jochem, P., Kinnear, N., et al. (2018). A review of consumer preferences of and interactions with electric vehicle charging infrastructure. *Transp. Res. Part D Transp. Environ.*, 62, 508–523. <https://doi.org/10.1016/j.trd.2018.04.002>.
- Higueras-Castillo, E., Singh, V., Singh, V., & Liébana-Cabanillas, F. (2023). Factors affecting adoption intention of electric vehicle: A cross-cultural study. *Environ. Dev. Sustain.*, 26(11), 29293–29329. <https://doi.org/10.1007/s10668-023-03865-y>.
- Jaiswal, D., Kaushal, V., Kant, R., & Singh, P. K. (2021). Consumer adoption intention for electric vehicles: Insights and evidence from Indian sustainable transportation. *Technol. Forecast. Soc. Change*, 173, 121089. <https://doi.org/10.1016/j.techfore.2021.121089>.
- Jansson, J., Nordlund, A., & Westin, K. (2017). Examining drivers of sustainable consumption: The influence of norms and opinion leadership on electric vehicle adoption in Sweden. *J. Cleaner Prod.*, 154, 176–187. <https://doi.org/10.1016/j.jclepro.2017.03.186>.
- Kumar, R. R. & Alok, K. (2020). Adoption of electric vehicle: A literature review and prospects for sustainability. *J. Cleaner Prod.*, 253, 119911. <https://doi.org/10.1016/j.jclepro.2019.119911>.
- Lashari, Z. A., Ko, J., & Jang, J. (2021). Consumers’ intention to purchase electric vehicles: Influences of user attitude and perception. *Sustainability*, 13(12), 6778. <https://doi.org/10.3390/su13126778>.
- Lazuardy, A., Nurcahyo, R., Farizal, F., & Oktavina, R. (2025). Formulating an electric vehicle adoption model in Indonesia: Product quality, consumer characteristics, adoption context, and perception. *Int. J. Technol.*, 16(6), 2122. <https://doi.org/10.14716/ijtech.v16i6.7804>.
- Li, W., Cui, K., Wu, L., & Zheng, B. (2025). Users’ perceived value of electric vehicles in China: A latent class model-based analysis. *World Electr. Veh. J.*, 16(8), 461. <https://doi.org/10.3390/wevj16080461>.
- Li, W., Long, R., & Chen, H. (2016). Consumers’ evaluation of national new energy vehicle policy in China: An analysis based on a four paradigm model. *Energy Policy*, 99, 33–41.

- <https://doi.org/10.1016/j.enpol.2016.09.050>.
- Luthra, S., Kumar, S., Garg, D., & Haleem, A. (2015). Barriers to renewable/sustainable energy technologies adoption: Indian perspective. *Renewable Sustainable Energy Rev.*, 41, 762–776. <https://doi.org/10.1016/j.rser.2014.08.077>.
- Megha (2024). Determinants of green consumption: A systematic literature review using the TCCM approach. *Front. Sustain.*, 5, 1428764. <https://doi.org/10.3389/frsus.2024.1428764>.
- Mesquita, A. R., de Abreu, V. H. S., Poyares, C. N., & Santos, A. S. (2025). Barriers to electric vehicle adoption: A framework to accelerate the transition to sustainable mobility. *Sustainability*, 17(18), 8318. <https://doi.org/10.3390/su17188318>.
- Mustafa, S., Shi, Y., Adan, D. e., Luo, W., & Al Humdan, E. (2026). Role of environmental awareness & self-identification expressiveness in electric-vehicle adoption. *Transportation*, 53(2), 861–885. <https://doi.org/10.1007/s11116-024-10515-3>.
- Noor, N. A. M., Muhammad, A., Isa, F. M., Shamsudin, M. F., & Abaidah, T. N. A. T. (2025). The electric vehicle (EV) revolution: How consumption values, consumer attitudes, and infrastructure readiness influence the intention to purchase electric vehicles in Malaysia. *World Electr. Veh. J.*, 16(10), 556. <https://doi.org/10.3390/wevj16100556>.
- Paul, J., Modi, A., & Patel, J. (2016). Predicting green product consumption using theory of planned behavior and reasoned action. *J. Retailing Consum. Serv.*, 29, 123–134. <https://doi.org/10.1016/j.jretconser.2015.11.006>.
- Pereira, S., Silva, C., & Ferreira, L. M. D. (2025). Barriers to energy transition: Comparing developing with developed countries. *Energy Sustainable Dev.*, 87, 101753. <https://doi.org/10.1016/j.esd.2025.101753>.
- Purwanto, E. & Irawan, A. P. (2024). The willingness to purchase electric vehicles: A comprehensive review. *IOP Conf. Ser.: Earth Environ. Sci.*, 1324, 012091. <https://doi.org/10.1088/1755-1315/1324/1/012091>.
- Purwanto, E., Uddin, N., & Nugraha, H. (2024). Hybrid solar-electric cart efficiency enhancement: A bibliometric analysis. *E3S Web Conf.*, 506, 11001. <https://doi.org/10.1051/e3sconf/202450611001>.
- Rahman, A., Suryawan, I. W. K., Suhardono, S., Nguyen, V. V., & Lee, C. (2025). Determinants of electric vehicle adoption in urban and peri-urban areas. *Energy Sustainable Dev.*, 85, 101664. <https://doi.org/10.1016/j.esd.2025.101664>.
- Rahmani, A. & Zamani, M. (2026). Explaining electric mobility technologies adoption: A meta-analysis of acceptance mechanisms across contexts. *Energy Convers. Manage.* X, 30, 101875. <https://doi.org/10.1016/j.ecmx.2026.101875>.
- Rezvani, Z., Jansson, J., & Bodin, J. (2015). Advances in consumer electric vehicle adoption research: A review and research agenda. *Transp. Res. Part D Transp. Environ.*, 34, 122–136. <https://doi.org/10.1016/j.trd.2014.10.010>.
- Salari, N. (2022). Electric vehicles adoption behaviour: Synthesising the technology readiness index with environmentalism values and instrumental attributes. *Transp. Res. Part A Policy Pract.*, 164, 60–81. <https://doi.org/10.1016/j.tra.2022.07.009>.
- Scheja, E. & Kim, K. B. (2024). Rethinking Economic Transformation for Sustainable and Inclusive Development: Turning a Corner? Edward Elgar Publishing Limited.
- Stockkamp, C., Schäfer, J., Millemann, J. A., & Heidenreich, S. (2021). Identifying factors associated with consumers' adoption of e-mobility—A systematic literature review. *Sustainability*, 13(19), 10975. <https://doi.org/10.3390/su131910975>.
- Tolani, K., Manohar, S., & Rao, S. (2025). Measuring sustainable mobility of electric vehicles: Determining critical factors with policy mix support for developing economies and user convenience. *J. Innovation Entrepreneurship*, 14(1), 1–35. <https://doi.org/10.1186/s13731-025-00560-2>.
- Uddin, N., Purwanto, E., & Nugraha, H. (2024). Machine learning based modeling for estimating solar power generation. *E3S Web Conf.*, 475, 03009. <https://doi.org/10.1051/e3sconf/202447503009>.
- Wang, S., Fan, J., Zhao, D., Yang, S., & Fu, Y. (2016). Predicting consumers' intention to adopt hybrid electric vehicles: Using an extended version of the theory of planned behavior model. *Transportation*, 43(1), 123–143. <https://doi.org/10.1007/s11116-014-9567-9>.
- Wu, D., Yu, L., Zhang, Q., Jiao, Y., & Wu, Y. (2021). Materialism, ecological consciousness and purchasing intention of electric vehicles: An empirical analysis among Chinese consumers. *Sustainability*, 13(5), 2964. <https://doi.org/10.3390/su13052964>.
- Yang, C., Qian, L., & Liu, M. (2026). Impact of government policies on the electric vehicle industry, environment, and society: A systematic review. *Technol. Soc.*, 85, 103226. <https://doi.org/10.1016/j.techsoc.2026.103226>.
- Zhao, H., Furuoka, F., AL Rasiyah, R., & Shen, E. (2024). Consumers' purchase intention toward electric vehicles from the perspective of perceived green value: An empirical survey from China. *World Electr. Veh. J.*, 15(6), 267. <https://doi.org/10.3390/wevj15060267>.